



(12) **United States Patent**
Tseng et al.

(10) **Patent No.:** **US 12,387,781 B2**
(45) **Date of Patent:** **Aug. 12, 2025**

(54) **CORRECTIVE READS WITH IMPROVED
RECOVERY FROM DATA RETENTION LOSS**

(71) Applicant: **Micron Technology, Inc.**, Boise, ID
(US)

(72) Inventors: **Huai-Yuan Tseng**, San Ramon, CA
(US); **Akira Goda**, Tokyo (JP);
Ching-Huang Lu, Fremont, CA (US);
Eric N. Lee, San Jose, CA (US);
Tomoharu Tanaka, Kanagawa (JP)

(73) Assignee: **Micron Technology, Inc.**, Boise, ID
(US)

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 198 days.

(21) Appl. No.: **18/142,112**

(22) Filed: **May 2, 2023**

(65) **Prior Publication Data**

US 2023/0360696 A1 Nov. 9, 2023

Related U.S. Application Data

(60) Provisional application No. 63/339,591, filed on May
9, 2022.

(51) **Int. Cl.**
G11C 11/4096 (2006.01)
G11C 11/408 (2006.01)
G11C 11/4093 (2006.01)

(52) **U.S. Cl.**
CPC **G11C 11/4096** (2013.01); **G11C 11/4085**
(2013.01); **G11C 11/4093** (2013.01)

(58) **Field of Classification Search**
CPC G11C 11/4096
USPC 365/185.09
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

11,513,887 B1 * 11/2022 Tishbi G06F 11/0751
2019/0043565 A1 * 2/2019 Miccoli G11C 11/5642
2022/0130466 A1 * 4/2022 Bazarsky G11C 16/3427

* cited by examiner

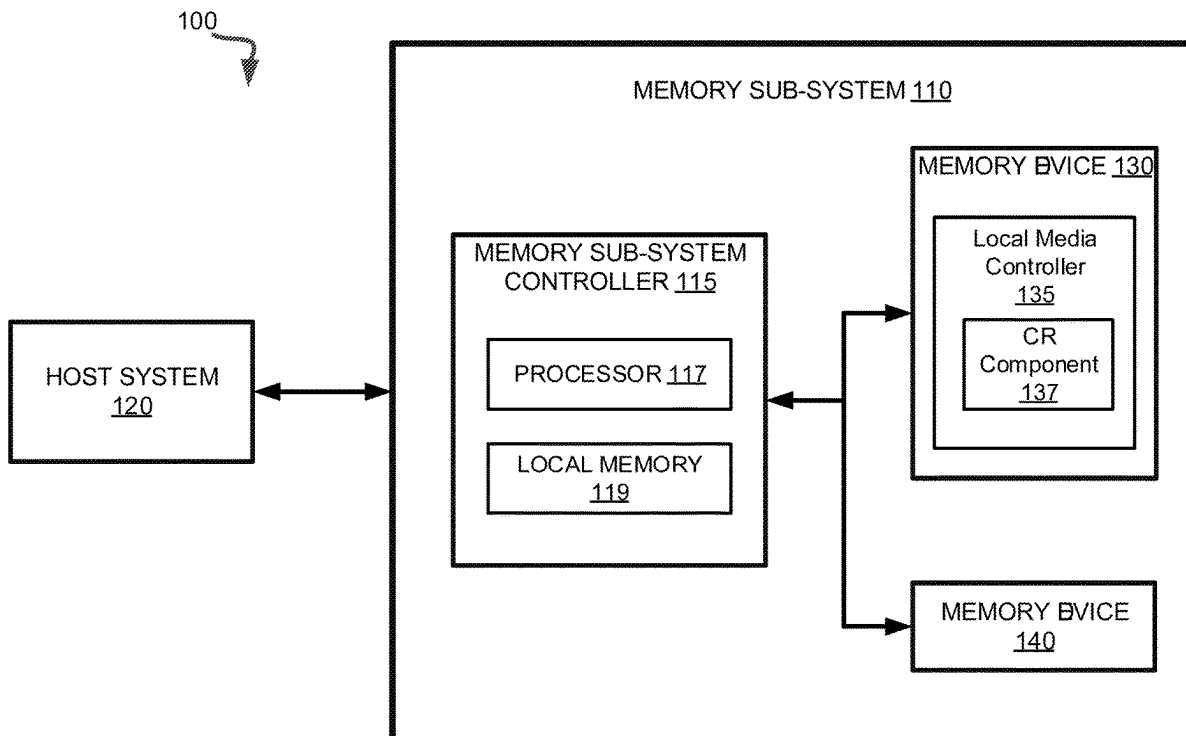
Primary Examiner — Anthan Tran

(74) *Attorney, Agent, or Firm* — Lowenstein Sandler LLP

(57) **ABSTRACT**

A read is initiated with respect to a target cell. A pair of adjacent cells includes a first cell and a second cell each adjacent to the target cell. First cell state information is obtained for the first cell and second cell state information is obtained for the second cell. A state information bin is determined by applying a pre-defined operation to the first cell state information and the second cell state information of the respective pair of adjacent cells. The target cell is assigned to the state information bin. Each state information bin defines a read level offset for reading the target cell.

20 Claims, 13 Drawing Sheets



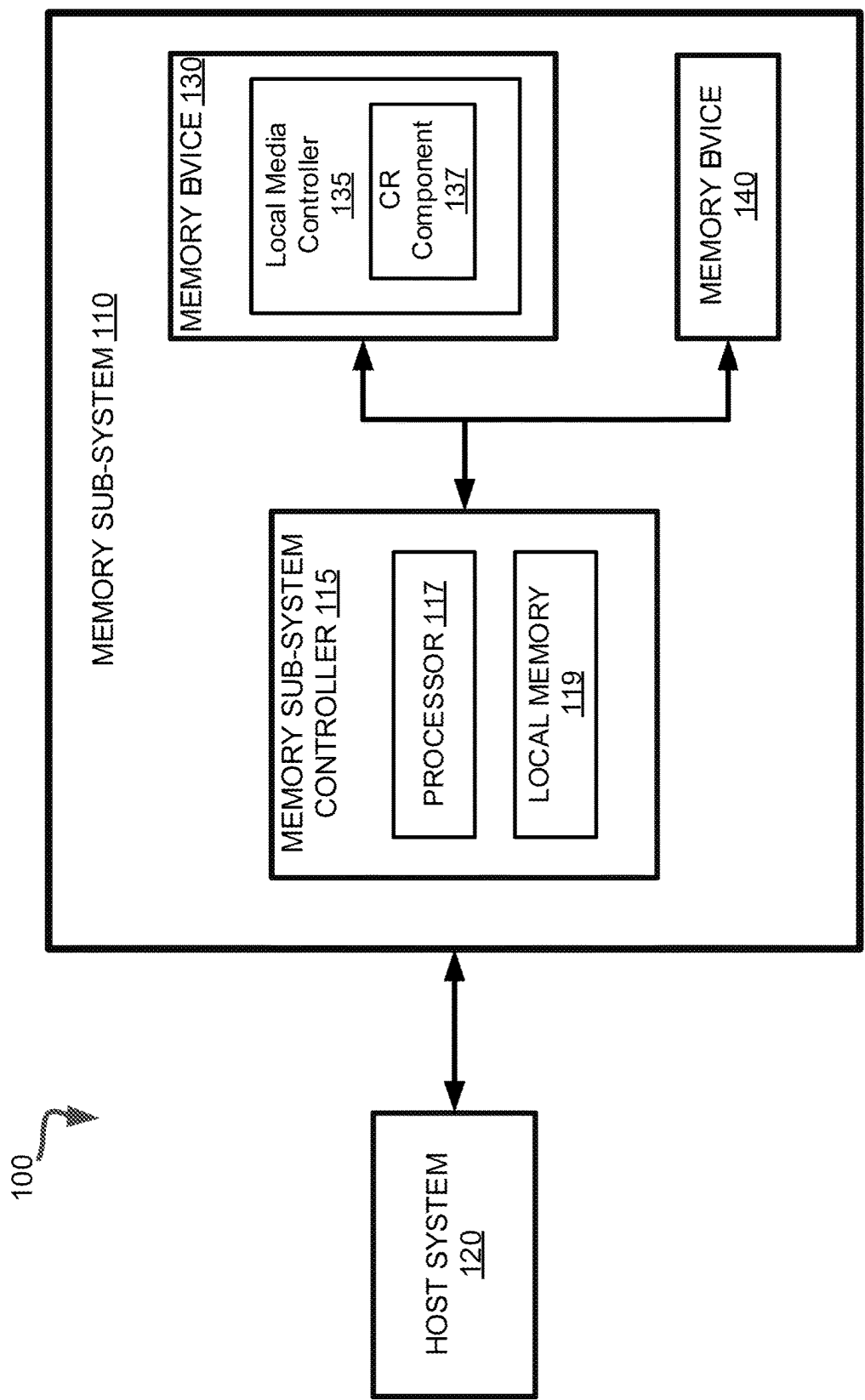


FIG. 1A

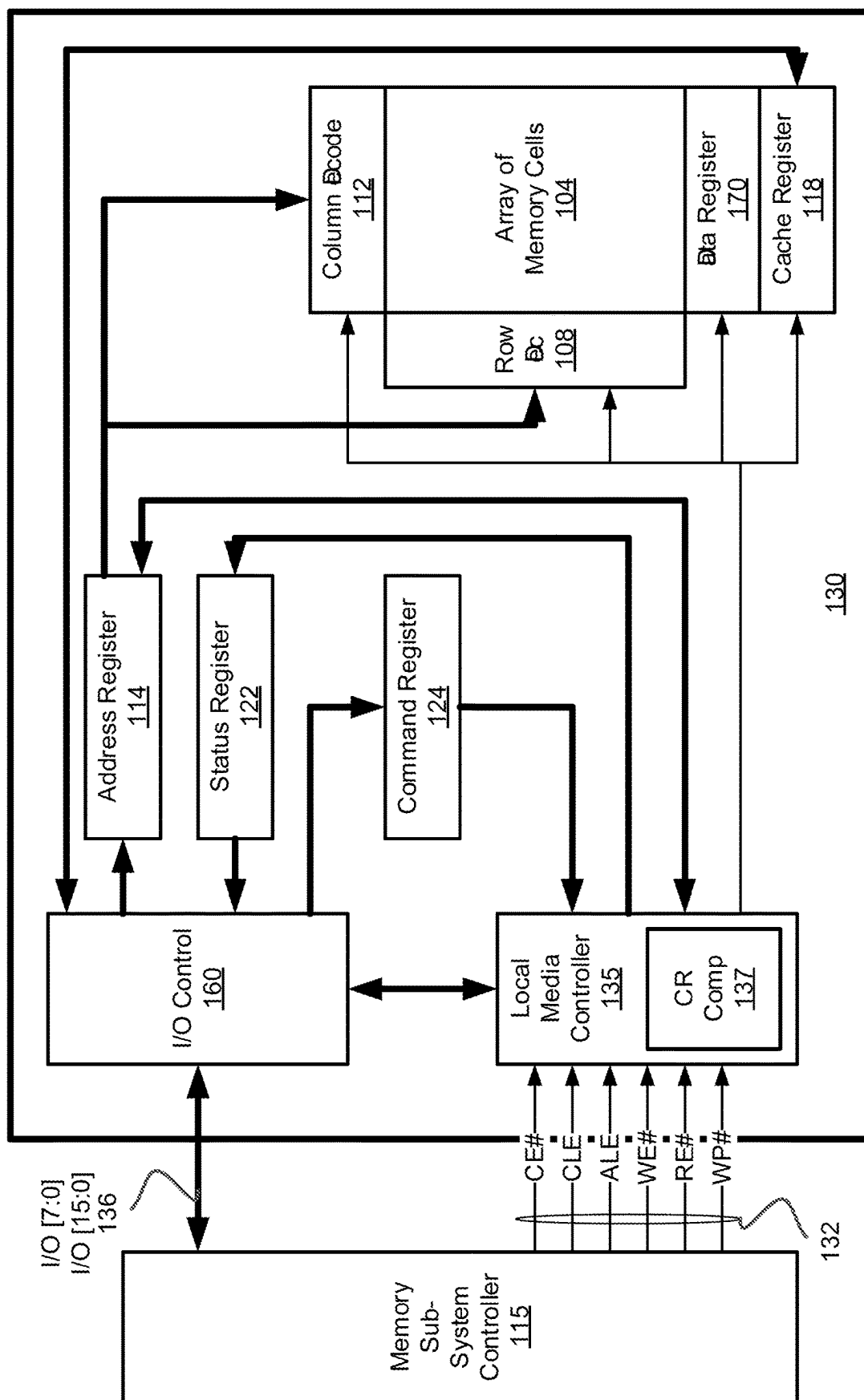


FIG. 1B

200A

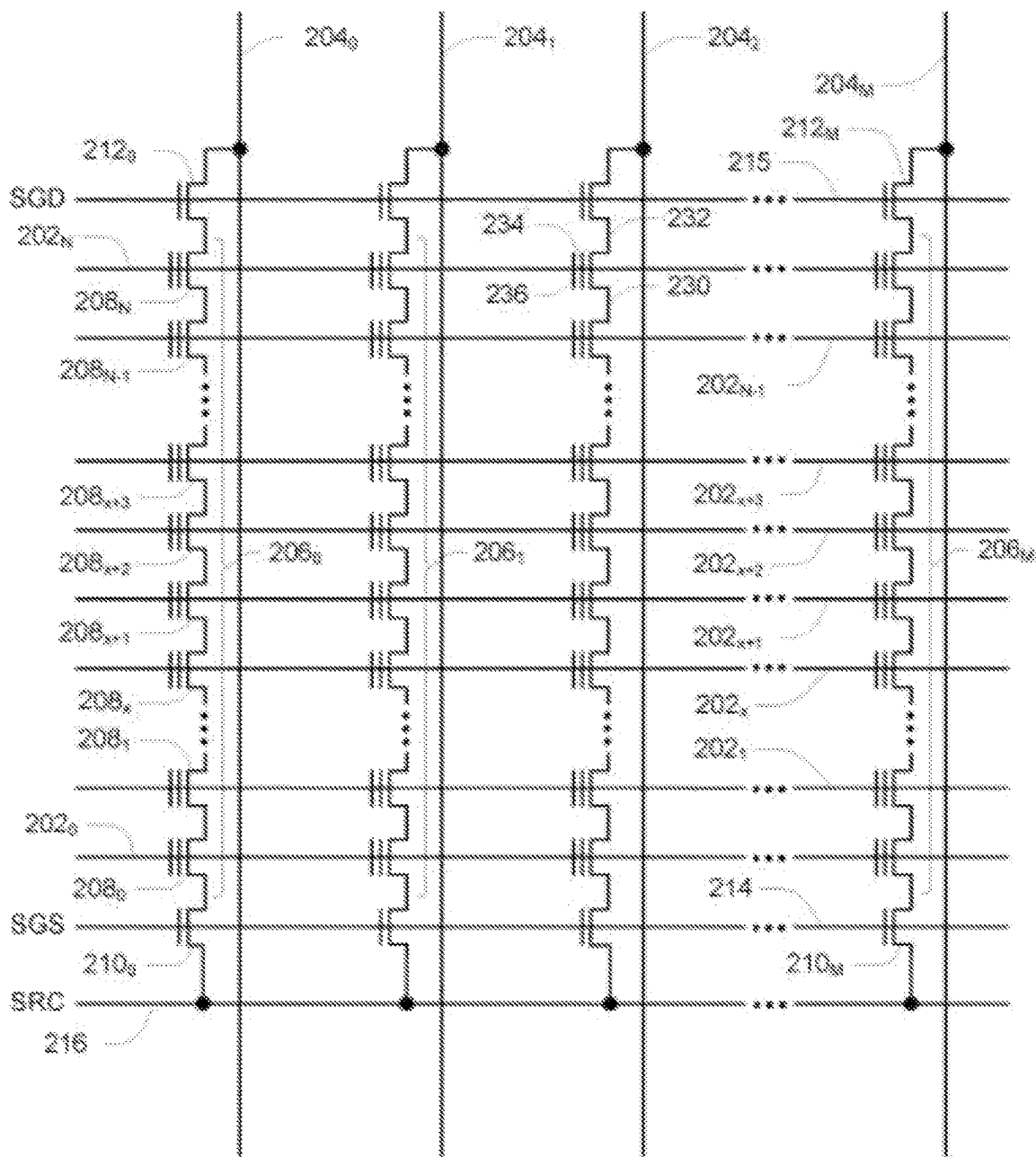



FIG. 2A

200B

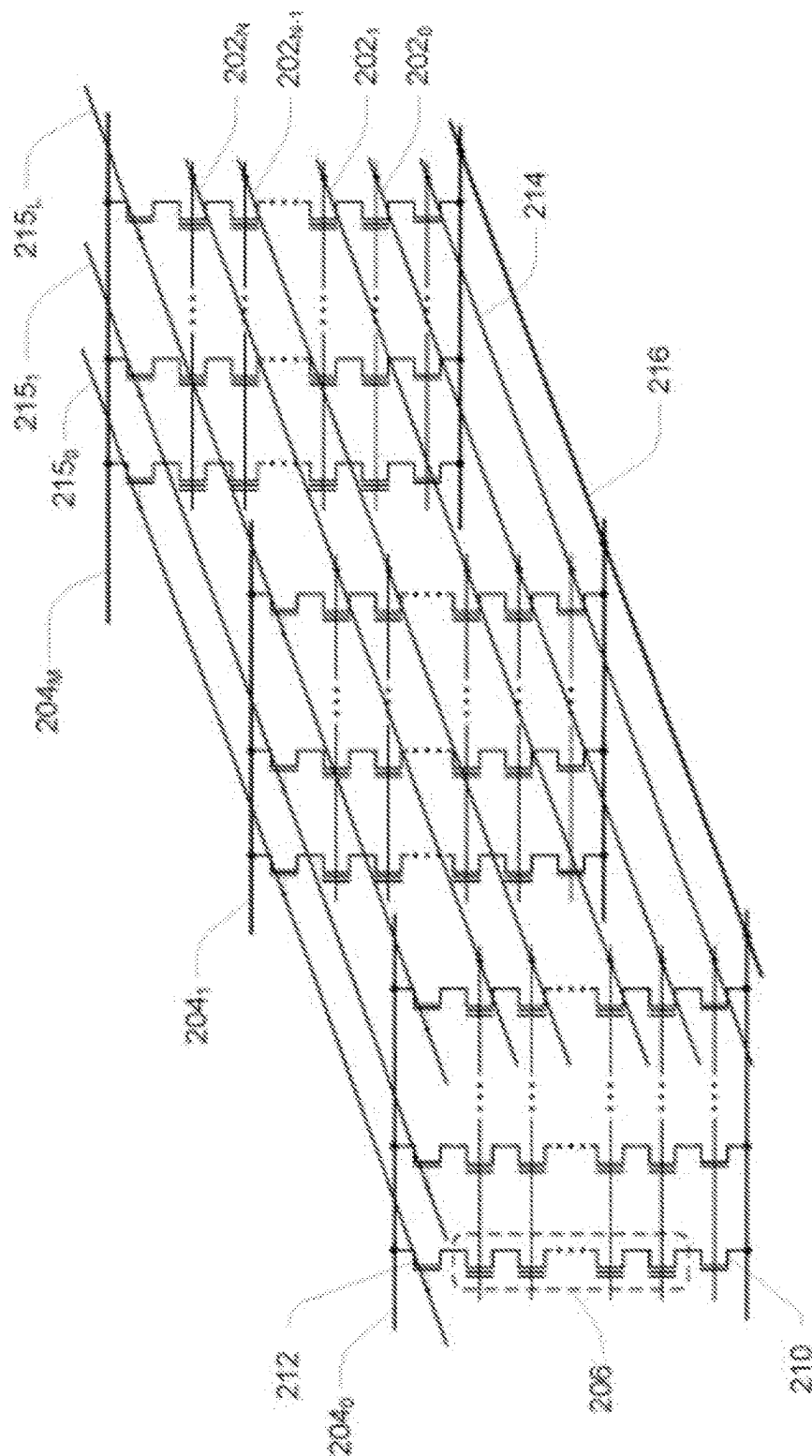


FIG. 2B

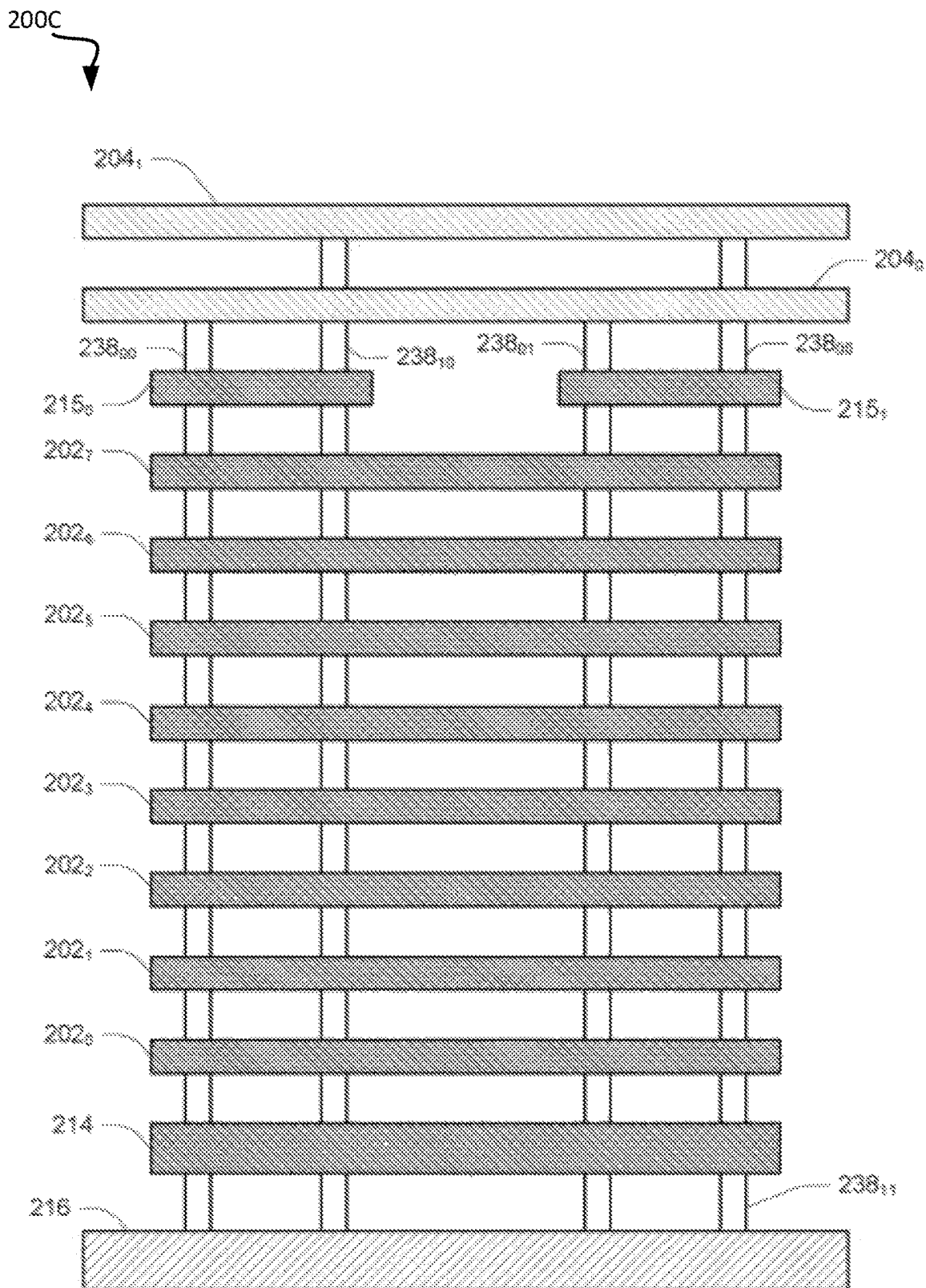


FIG. 2C

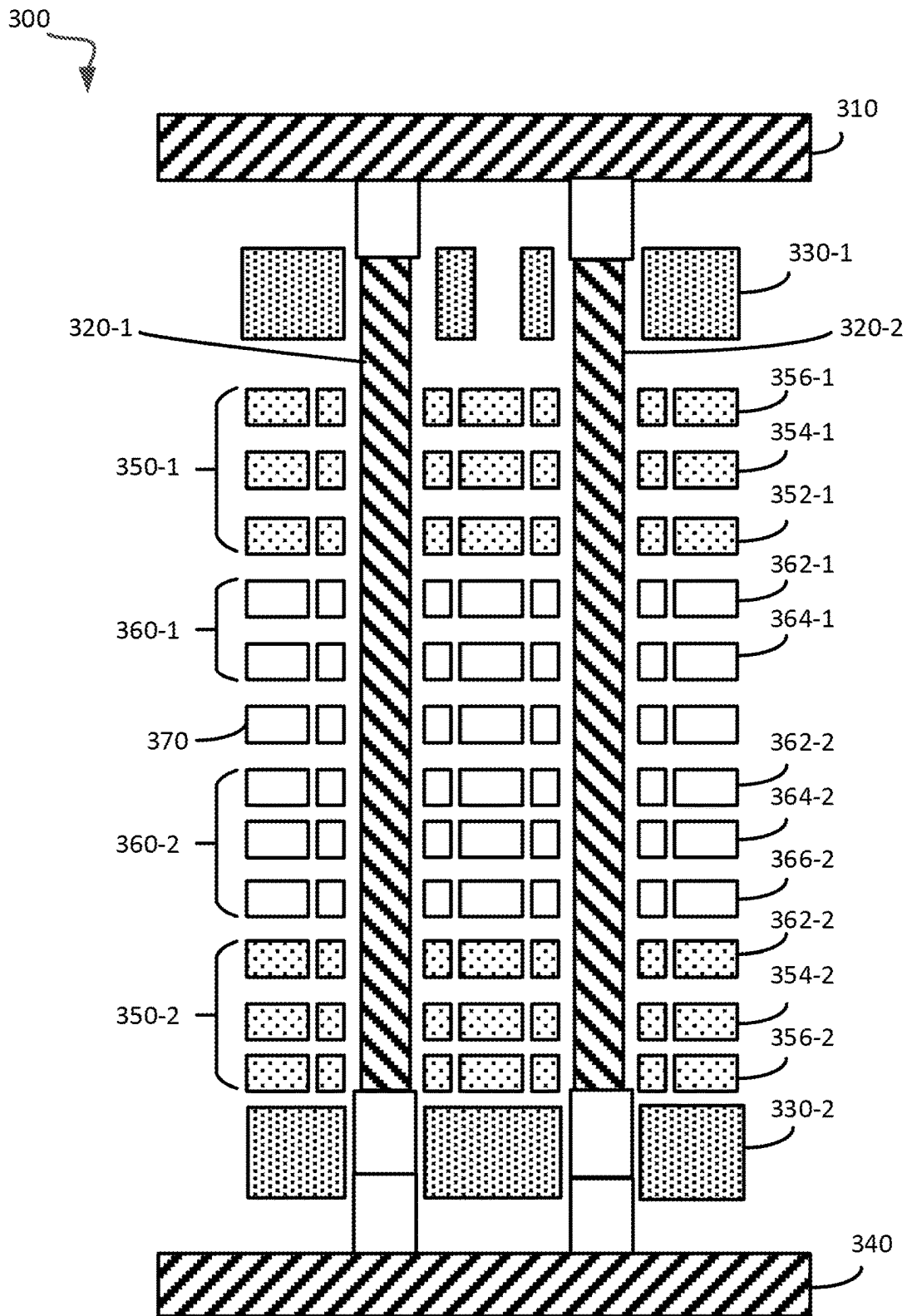


FIG. 3

400

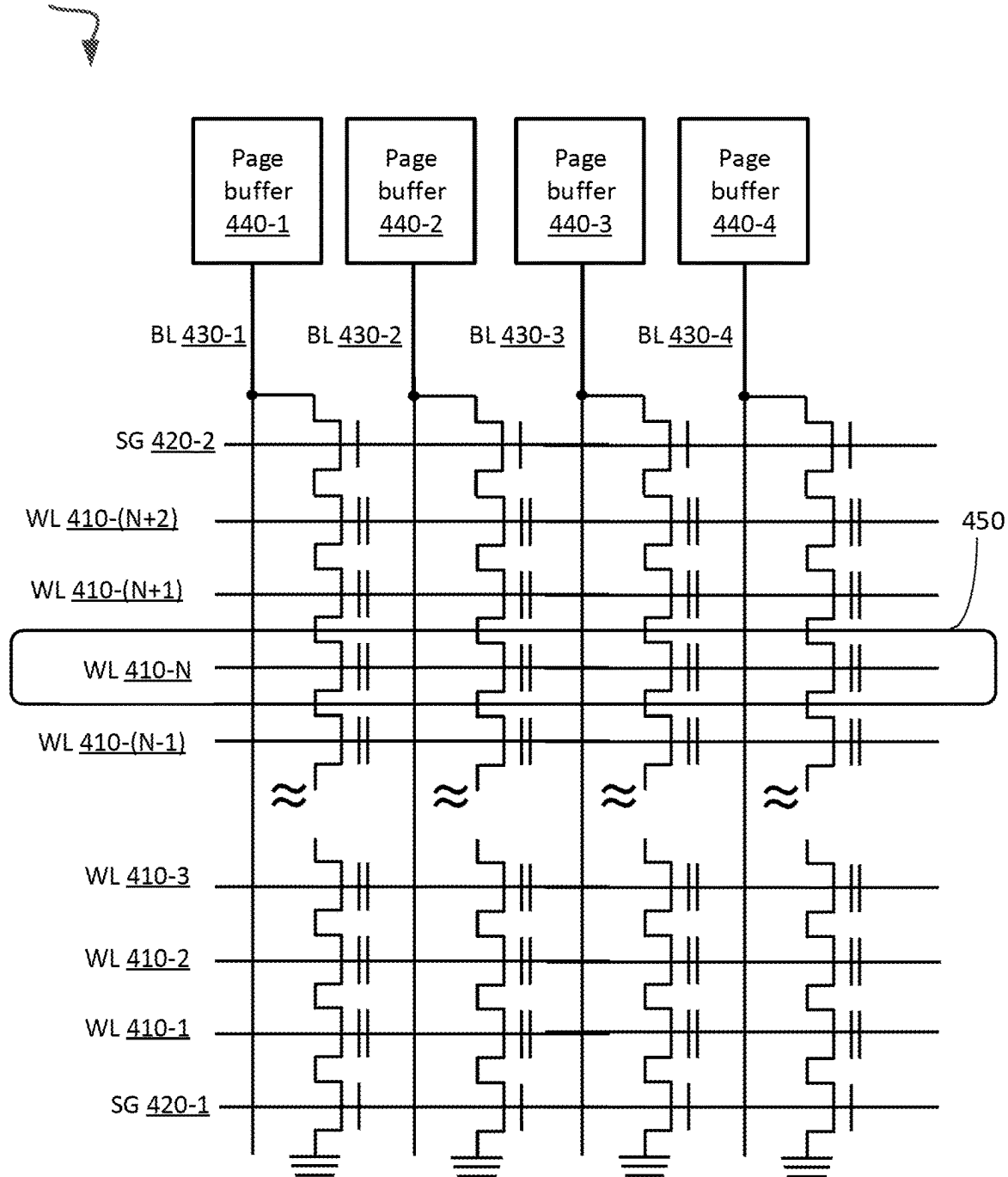


FIG. 4

500

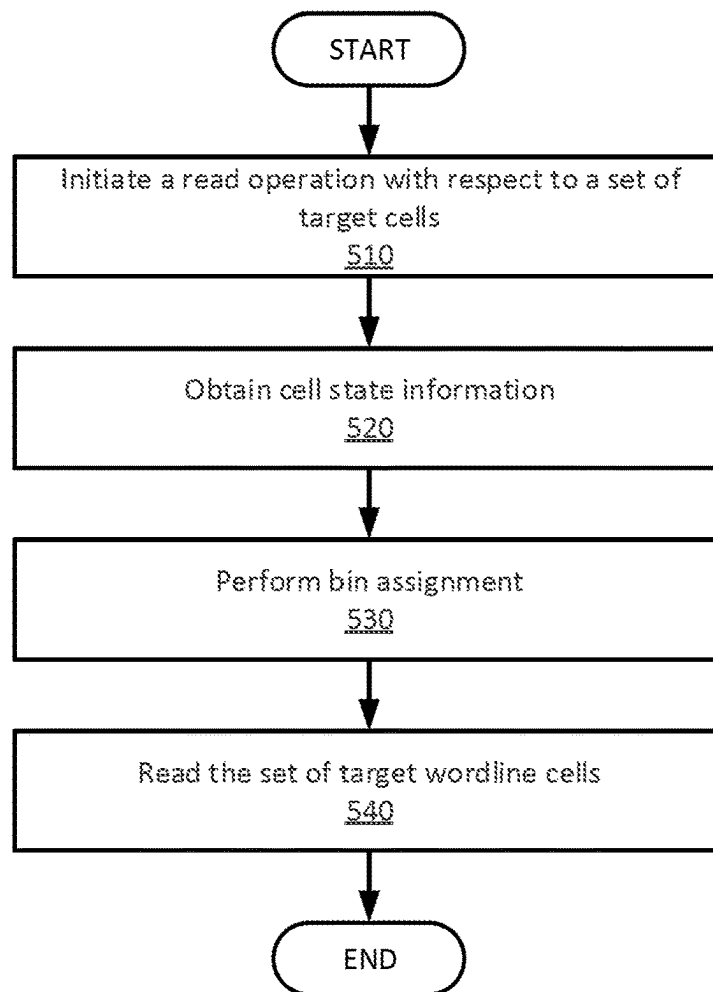


FIG. 5

600B



WLn+1	4	L15	4	4	4	5	5	5	6	6	6	7	7	7	8	8	8	8
	4	L14	4	4	4	5	5	5	6	6	6	7	7	7	8	8	8	8
	4	L13	4	4	4	5	5	5	6	6	6	7	7	7	8	8	8	8
	4	L12	4	4	4	5	5	5	6	6	6	7	7	7	8	8	8	8
	3	L11	3	3	3	4	4	4	5	5	5	6	6	6	7	7	7	7
	3	L10	3	3	3	4	4	4	5	5	5	6	6	6	7	7	7	7
	3	L9	3	3	3	4	4	4	5	5	5	6	6	6	7	7	7	7
	2	L8	2	2	2	3	3	3	4	4	4	5	5	5	6	6	6	6
	2	L7	2	2	2	3	3	3	4	4	4	5	5	5	6	6	6	6
	2	L6	2	2	2	3	3	3	4	4	4	5	5	5	6	6	6	6
	1	L5	1	1	1	2	2	2	3	3	3	4	4	4	5	5	5	5
	1	L4	1	1	1	2	2	2	3	3	3	4	4	4	5	5	5	5
	1	L3	1	1	1	2	2	2	3	3	3	4	4	4	5	5	5	5
	0	L2	0	0	0	1	1	1	2	2	2	3	3	3	4	4	4	4
	0	L1	0	0	0	1	1	1	2	2	2	3	3	3	4	4	4	4
0	L0	0	0	0	1	1	1	2	2	2	3	3	3	4	4	4	4	
WLn-1 bit	L0	L1	L2	L3	L4	L5	L6	L7	L8	L9	L10	L11	L12	L13	L14	L15		
	WLn+1 bit	0	0	0	1	1	2	2	2	3	3	3	4	4	4	4	4	
		WLn+1																

FIG. 6B

600C

I-ETM	L15	5	5	6	6	7	7	7	8	8	8	9	9	9	10	10
	L14	5	5	6	6	7	7	7	8	8	8	9	9	9	10	10
	L13	4	4	5	5	6	6	6	7	7	7	8	8	8	9	9
	L12	4	4	5	5	6	6	6	7	7	7	8	8	8	9	9
	L11	4	4	5	5	6	6	6	7	7	7	8	8	8	9	9
	L10	3	3	4	4	5	5	5	6	6	6	7	7	7	8	8
	L9	3	3	4	4	5	5	5	6	6	6	7	7	7	8	8
	L8	3	3	4	4	5	5	5	6	6	6	7	7	7	8	8
	L7	2	2	3	3	4	4	4	5	5	5	6	6	6	7	7
	L6	2	2	3	3	4	4	4	5	5	5	6	6	6	7	7
	L5	2	2	3	3	4	4	4	5	5	5	6	6	6	7	7
	L4	1	1	2	2	3	3	3	4	4	4	5	5	5	6	6
	L3	1	1	2	2	3	3	3	4	4	4	5	5	5	6	6
	L2	1	1	2	2	3	3	3	4	4	4	5	5	5	6	6
	L1	0	0	1	1	2	2	2	3	3	3	4	4	4	5	5
	L0	0	0	1	1	2	2	2	3	3	3	4	4	4	5	5
WLn-1bn	L0	L1	L2	L3	L4	L5	L6	L7	L8	L9	L10	L11	L12	L13	L14	L15
WLn+1bn	0	0	1	1	2	2	2	3	3	3	4	4	4	5	5	5
	WLn+1															

FIG. 6C

600D

I-EM	6	L15	6	6	7	7	8	8	9	9	9	10	10	11	11	12	12
	6	L14	6	6	7	7	8	8	9	9	9	10	10	11	11	12	12
	5	L13	5	5	6	6	7	7	8	8	8	9	9	10	10	11	11
	5	L12	5	5	6	6	7	7	8	8	8	9	9	10	10	11	11
	4	L11	4	4	5	5	6	6	7	7	7	8	8	9	9	10	10
	4	L10	4	4	5	5	6	6	7	7	7	8	8	9	9	10	10
	3	L9	3	3	4	4	5	5	6	6	6	7	7	8	8	9	9
	3	L8	3	3	4	4	5	5	6	6	6	7	7	8	8	9	9
	3	L7	3	3	4	4	5	5	6	6	6	7	7	8	8	9	9
	2	L6	2	2	3	3	4	4	5	5	5	6	6	7	7	8	8
	2	L5	2	2	3	3	4	4	5	5	5	6	6	7	7	8	8
	2	L4	2	2	3	3	4	4	5	5	5	6	6	7	7	8	8
	1	L3	1	1	2	2	3	3	4	4	4	5	5	6	6	7	7
	1	L2	1	1	2	2	3	3	4	4	4	5	5	6	6	7	7
	0	L1	0	0	1	1	2	2	3	3	3	4	4	5	5	6	6
	0	L0	0	0	1	1	2	2	3	3	3	4	4	5	5	6	6
	WLn-1 bin	L0	L1	L2	L3	L4	L5	L6	L7	L8	L9	L10	L11	L12	L13	L14	L15
	WLn+1 bin	0	0	1	1	2	2	2	3	3	3	4	4	5	5	6	6
		WLn+1															

FIG. 6D

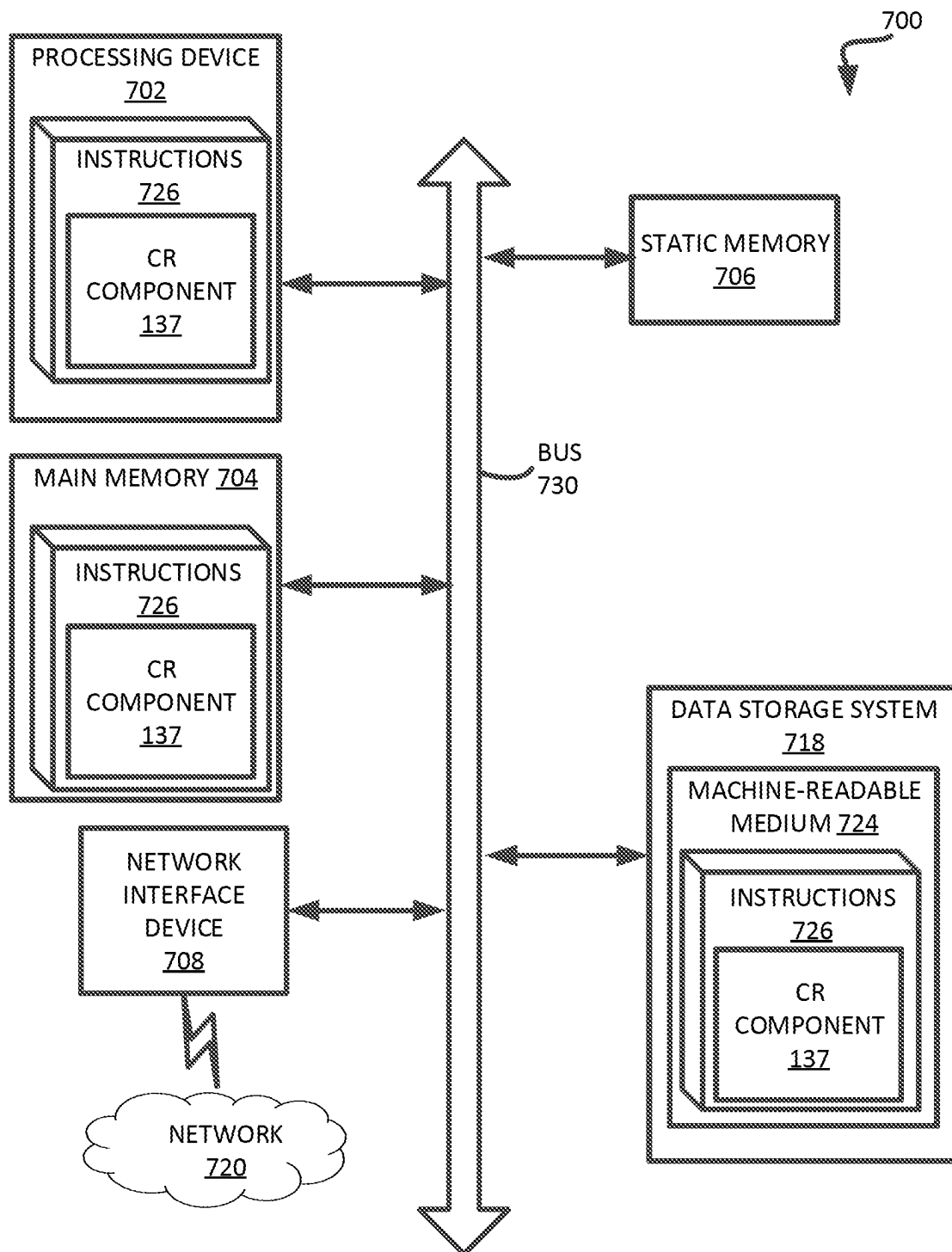


FIG. 7

1

CORRECTIVE READS WITH IMPROVED RECOVERY FROM DATA RETENTION LOSS

RELATED APPLICATION

This application claims the benefit of U.S. Provisional Application 63/339,591, filed on May 9, 2022 and entitled “CORRECTIVE READS WITH IMPROVED RECOVERY FROM DATA RETENTION LOSS”, the entire contents of which are incorporated by reference herein.

TECHNICAL FIELD

Embodiments of the disclosure relate generally to memory sub-systems, and more specifically, relate to performing corrective reads with improved recovery from data retention loss.

BACKGROUND

A memory sub-system can include one or more memory devices that store data. The memory devices can be, for example, non-volatile memory devices and volatile memory devices. In general, a host system can utilize a memory sub-system to store data at the memory devices and to retrieve data from the memory devices.

BRIEF DESCRIPTION OF THE DRAWINGS

The disclosure will be understood more fully from the detailed description given below and from the accompanying drawings of various embodiments of the disclosure. The drawings, however, should not be taken to limit the disclosure to the specific embodiments, but are for explanation and understanding only.

FIG. 1 illustrates an example computing system that includes a memory sub-system in accordance with some embodiments of the present disclosure.

FIG. 1B is a block diagram of a memory device in communication with a memory sub-system controller of a memory sub-system in accordance with some embodiments of the present disclosure.

FIGS. 2A-2C are diagrams of portions of an example array of memory cells included in a memory device, in accordance with some embodiments of the present disclosure.

FIG. 3 is a diagram of an example three-dimensional (3D) replacement gate memory device that can enable performance of corrective reads implementing incremental reads, in accordance with some embodiments of the present disclosure.

FIG. 4 is a diagram of a portion of a memory device, in accordance with some embodiments of the present disclosure.

FIG. 5 is a flow diagram of an example method to perform a corrective read with improved recovery from data retention loss, in accordance with some embodiments of the present disclosure.

FIGS. 6A-6D are tables of example sets of bins that can be used to perform a corrective read with improved recovery from data retention loss, in accordance with some embodiments of the present disclosure.

FIG. 7 is a block diagram of an example computer system in which embodiments of the present disclosure may operate.

DETAILED DESCRIPTION

Aspects of the present disclosure are directed to corrective reads implementing efficient binning. A memory sub-system

2

can be a storage device, a memory module, or a combination of a storage device and memory module. Examples of storage devices and memory modules are described below in conjunction with FIGS. 1A-1B. In general, a host system can utilize a memory sub-system that includes one or more components, such as memory devices that store data. The host system can provide data to be stored at the memory sub-system and can request data to be retrieved from the memory sub-system.

A memory sub-system can include high density non-volatile memory devices where retention of data is desired when no power is supplied to the memory device. One example of non-volatile memory devices is a negative-and (NAND) memory device. Other examples of non-volatile memory devices are described below in conjunction with FIGS. 1A-1B. A non-volatile memory device is a package of one or more dies. Each die includes one or more planes. For some types of non-volatile memory devices (e.g., NAND devices), each plane includes a set of physical blocks. Each block consists of a set of pages. Each page includes a set of memory cells. A memory cell is an electronic circuit that stores information. Depending on the memory cell type, a memory cell can store one or more bits of binary information, and has various logic states that correlate to the number of bits being stored. The logic states can be represented by binary values, such as “0” and “1”, or combinations of such values.

A memory device can include multiple memory cells arranged in a two-dimensional or three-dimensional grid. Memory cells are formed onto a silicon wafer in an array of columns and rows. A memory device can further include conductive lines connected to respective ones of the memory cells, referred to as wordlines and bitlines. A wordline can refer to one or more rows of memory cells of the memory device and a bitline can refer to one or more columns of memory cells. The intersection of a bitline and wordline constitutes the address of the memory cell. A block herein-after refers to a unit of the memory device used to store data and can include a group of memory cells, a wordline group, a wordline, or individual memory cells. One or more blocks can be grouped together to form a plane of the memory device in order to allow concurrent operations to take place on each plane. The memory device can include circuitry that performs concurrent memory page accesses of two or more memory planes. For example, the memory device can include a respective access line driver circuit and power circuit for each plane of the memory device to facilitate concurrent access of pages of two or more memory planes, including different page types.

Some memory devices can be three-dimensional (3D) memory devices (e.g., 3D NAND devices). For example, a 3D memory device can include memory cells that are placed between sets of layers including a pillar (e.g., polysilicon pillar), a tunnel oxide layer, a charge trap (CT) layer, and a dielectric (e.g. oxide) layer. A 3D memory device can have a “top deck” corresponding to a first side and a “bottom deck” corresponding to a second side. Without loss of generality, the first side can be a drain side and the second side can be a source side. For example, a 3D memory device can be a 3D replacement gate memory device having a replacement gate structure using wordline stacking.

A memory cell (“cell”) can be programmed (written to) by applying a certain voltage to the cell, which results in an electric charge being held by the cell. For example, a voltage signal V_{CG} that can be applied to a control electrode of the cell to open the cell to the flow of electric current across the cell, between a source electrode and a drain electrode. More

specifically, for each individual cell (having a charge Q stored thereon) there can be a threshold control gate voltage V_T (also referred to as the “threshold voltage”) such that the source-drain electric current is low for the control gate voltage (V_{CG}) being below the threshold voltage, $V_{CG} < V_T$. The current increases substantially once the control gate voltage has exceeded the threshold voltage, $V_{CG} > V_T$. Because the actual geometry of the electrodes and gates varies from cell to cell, the threshold voltages can be different even for cells implemented on the same die. The cells can, therefore, be characterized by a distribution P of the threshold voltages, $P(Q, V_T) = dW/dV_T$, where dW represents the probability that any given cell has its threshold voltage within the interval $[V_T, V_T + dV_T]$ when charge Q is placed on the cell.

A memory device can exhibit threshold voltage distributions $P(Q, V_T)$ that are narrow compared with the working range of control voltages tolerated by the cells of the device. Accordingly, multiple non-overlapping distributions $P(Q_k, V_T)$ (“valleys”) can be fit into the working range allowing for storage and reliable detection of multiple values of the charge Q_k , $k=1, 2, 3, \dots$. The distributions (valleys) are interspersed with voltage intervals (“valley margins”) where none (or very few) of the cells of the device have their threshold voltages. Such valley margins can, therefore, be used to separate various charge states Q_k —the logical state of the cell can be determined by detecting, during a read operation, between which two valley margins the respective threshold voltage V_T of the cell resides. Specifically, the read operation can be performed by comparing the measured threshold voltage V_T exhibited by the memory cell to one or more reference voltage levels corresponding to known valley margins (e.g., centers of the margins) of the memory device.

One type of cell is a single level cell (SLC), which stores 1 bit per cell and defines 2 logical states (“states”) (“1” or “L0” and “0” or “L1”) each corresponding to a respective V_T level. For example, the “1” state can be an erased state and the “0” state can be a programmed state (L1). Another type of cell is a multi-level cell (MLC), which stores 2 bits per cell (1 bit for upper page (UP) data and 1 bit for lower page (LP) data) and defines 4 states (“11” or “L0”, “10” or “L1”, “01” or “L2” and “00” or “L3”) each corresponding to a respective V_T level. For example, the “11” state can be an erased state and the “01”, “10” and “00” states can each be a respective programmed state. Another type of cell is a triple level cell (TLC), which stores 3 bits per cell (1 bit for UP data, 1 bit for LP data and 1 bit for extra page (XP) data) and defines 8 states (“111” or “L0”, “110” or “L1”, “101” or “L2”, “100” or “L3”, “011” or “L4”, “010” or “L5”, “001” or “L6”, and “000” or “L7”) each corresponding to a respective V_T level. For example, the “111” state can be an erased state and each of the other states can be a respective programmed state. Another type of a cell is a quad-level cell (QLC), which stores 4 bits per cell (1 bit for UP data, 1 bit for LP data, 1 bit for XP data, and 1 bit for top page (TP) data) and defines 16 states L0-L15, where L0 corresponds to “1111” and L15 corresponds to “0000”. Another type of cell is a penta-level cell (PLC), which stores 5 bits per cell and defines 32 states. Other types of cells are also contemplated. Thus, an n -level cell can use 2^n levels of charge to store n bits of information for n pages. A memory device can include one or more arrays of memory cells such as SLCs, MLCs, TLCs, QLCs, PLCs, etc. or any combination of such. For example, a memory device can include an SLC portion, and an MLC portion, a TLC portion, a QLC portion, or a PLC portion of cells.

A valley margin can also be referred to as a read window. For example, in a SLC cell, there is 1 read window that exists with respect to the 2 V_T distributions. As another example, in an MLC cell, there are 3 read windows that exist with respect to the 4 V_T distributions. As yet another example, in a TLC cell, there are 7 read windows that exist with respect to the 8 V_T distributions. Read window size generally decreases as the number of states increases. For example, the 1 read window for the SLC cell may be larger than each of the 3 read windows for the MLC cell, and each of the 3 read windows for the MLC cell may be larger than each of the 7 read windows for the TLC cell, etc. Read window budget (RWB) refers to the cumulative value of the read windows.

Cells of a memory array that are selected to be read during a read operation can be referred to target cells connected to a target wordline. The target cells can neighbor adjacent cells connected to at least one wordline neighboring the target wordline (“adjacent wordline”). For example, the at least one adjacent wordline can be a single wordline neighboring the target wordline or a pair of wordlines neighboring the target wordline. Illustratively, the target wordline can be referred to as an n -th wordline (WL_n), and the at least one adjacent wordline can include at least one of adjacent wordline $n-1$ (WL_{n-1}) or adjacent wordline $n+1$ (WL_{n+1}). For example, in a 3D memory device, the set of adjacent wordlines can include a wordline located directly above the target wordline and/or a wordline located directly below the target wordline.

Each target cell has a respective group of adjacent cells. Each group of adjacent cells includes at least one cell that neighbors its respective target cell (e.g., one cell connected to WL_{n-1} and/or one cell connected to WL_{n+1}). More specifically, each target cell can be connected to the same bitline as each cell of the respective group of adjacent cells, such that the target cell and the cells of the respective group of adjacent cells are within the same string. Accordingly, each group of adjacent cells can include a single adjacent cell, or a pair of adjacent cells connected to a same bitline as a respective target cell.

A read can include a prologue phase during which a controller activates voltage pumps (e.g., causes voltage pumps to be turned on) and loads information for the read operation, a strobe phase in which a number of strobes are performed, and an epilogue phase during which the controller causes the cells to discharge, deactivates the voltage pumps (e.g., causes the voltage pumps to be turned off) and causes the memory device to return to an idle or standby state (e.g., depending on the state of the CE #signal). A strobe refers to a read performed at a particular read level offset. For example, for a 3 strobe page type, a 3 strobe read can be performed during the strobe phase.

Cell-to-cell interference may exist in a memory array between the target cells and their respective groups of adjacent cells. Cell-to-cell interference can lead to lateral charge migration and V_T distribution shift. Cell-to-cell interference, in addition to intrinsic charge loss, can further lead to a widening of V_T distributions. The V_T distribution widening can cause RWB degradation, which can negatively affect memory device reliability. For example, RWB degradation can lead to an increase in the number of errors (e.g., bit errors) and/or error rate (e.g., bit error rate (BER)).

One mechanism to compensate for the effects of cell-to-cell interference and/or intrinsic charge loss is corrective read. Generally, a corrective read operation is performed to read each target cell using an appropriate read level offset that accounts for the cell-to-cell inference, lateral charge

migration and/or intrinsic charge loss caused by the respective group of adjacent cells. The read level offset can be applied with respect to a center read level. For example, the center read level can be located within a valley between target cell V_T distributions.

To implement a corrective read operation, a controller can, for each group of adjacent cells, obtain cell state information for each cell of the group of adjacent cells. The cell state information for a cell reflects the logical level (e.g., L0-Ln, where n is the total number of logical levels supported) of the cell. For example, if a cell is an SLC cell, the cell state information can reflect whether the cell is in the L0 state or the L1 state. As another example, if the cell is a TLC cell, the cell state information can reflect which of the states L0-L7 that the cell is in. The cell state information for a cell can be obtained by identifying the state of the cell.

To identify the state of the cell, the controller can cause a read voltage to be applied the cell (e.g., gate electrode of the cell) and determine whether the read voltage activates (e.g., turns on) the cell. If the read voltage activates the cell, this indicates that the read voltage is greater than or equal to the V_T of the cell. Additional read voltage(s) may be applied to the cell to determine whether the cell is in a lower state. If the read voltage does not activate the cell, this means that the read voltage is less than the V_T of the cell, and that the cell is in a higher state. Additional read voltage(s) may be applied until the cell is activated. For each group of adjacent cells, the controller can store the cell state information for each cell of the group of adjacent cells in a respective page buffer (e.g., static page buffer). Each page buffer can be connected to a respective group of adjacent cells via a bitline.

In some embodiments, the cell state information for each cell of a group of adjacent cells is 1-bit information. For example, obtaining the 1-bit cell state information can involve applying a single strobe read to each cell of the group of adjacent cells. A strobe refers to a read performed at a particular read level offset. If the group of adjacent cells includes a single cell (e.g., a cell connected to one of the adjacent wordlines WL_{n-1} and WL_{n+1}), then the stored cell state information is 1 bit in total. The 1 bit stored cell state information can be used to implement 1-bit corrective read (1BCR). If the group of adjacent cells includes a pair of cells (e.g., cells connected to the adjacent wordlines WL_{n-1} and WL_{n+1}), then the stored cell state information is 2 bits in total. The 2 bit stored cell state information can be used to implement a "1-bit 2-sided" version of 2-bit corrective read (2BCR).

In some embodiments, the cell state information for each cell of a group of adjacent cells is 2-bit information. For example, obtaining the 2-bit cell state information can involve applying a three strobe read to each cell of the group of adjacent cells. If the group of adjacent cells includes a single adjacent cell (e.g., a cell connected to one of the adjacent wordlines WL_{n-1} and WL_{n+1}), then the stored cell state information is 2 bits in total. The 2 bit stored cell state information can be used to implement a "2-bit 1-sided" version of 2BCR. If the group of adjacent cells includes a pair of cells (e.g., cells connected to the adjacent wordlines WL_{n-1} and WL_{n+1}), then the stored cell state information is 4 bits in total. The 4 bit stored cell state information can be used to implement 4-bit corrective read (4BCR).

The controller can then assign each target cell to a respective state information bin ("bin") using the cell state information for each cell of the respective group of adjacent cells. As will be described in further detail below, each bin defines a respective cell state information condition, and a

target cell is assigned to a bin if the cell(s) within its respective group of adjacent cells satisfy the cell state information condition for the bin. Thus, the bins divide each target cell V_T distribution into a number of respective target cell V_T sub-distributions, where each target cell V_T sub-distribution is formed from the target cells assigned to a respective bin.

For example, assume that a group of adjacent wordline cells for a target cell connected to WL_n is a single adjacent cell connected to WL_{n-1} or WL_{n+1} . The cell state information condition for a bin can define a range of states for single adjacent cells. A target cell can then be assigned to the bin if the cell state information for the single adjacent cell indicates that the state of the single adjacent cell falls within the range.

As another example, assume that a group of adjacent wordline cells for a target cell connected to WL_n is a pair of adjacent cells including a first adjacent cell connected to WL_{n-1} and a second adjacent cell connected to WL_{n+1} . The cell state information condition for a bin can define a first range of states for first adjacent cells and a second range of states for second adjacent cells. The target cell can then be assigned to the bin if the cell state information for the first adjacent cell indicates that the state of the first adjacent cell falls within the first range and if the cell state information for the second adjacent cell indicates that the state of the second adjacent cell falls within the second range.

The number of bins can be determined by the particular corrective read implementation (e.g., 1BCR, 2BCR or 4BCR). More specifically, the number of bins can be determined as 2^B , where B equals the total number of bits of cell state information stored for a group of adjacent cells.

For example, in a 1BCR implementation in which $B=1$, the number of bins is 2 (i.e., 2^1) and each bin defines a respective cell state information condition. Illustratively, for a QLC cell implementation in which there are 16 total possible states L0-L15, a target cell can be assigned to a first bin assigned with a first read level offset if its single adjacent cell is determined to have a state from L0-L7, and a target cell can be assigned to a second bin assigned with a second read level offset if its single adjacent cell is determined to have a state from L8-L15.

As another example, in a 2-bit 1-sided 2BCR implementation in which $B=2$, the number of bins is 4 (i.e., 2^2) and each bin defines a respective cell state information condition. Illustratively, for the QLC cell implementation, a target cell can be assigned to a first bin assigned with a first read level offset if its single adjacent cell is determined to have a state from L0-L3, a target cell can be assigned to a second bin if its single adjacent cell is determined to have a state from L4-L7, a target cell can be assigned to a third bin if its single adjacent cell is determined to have a state from L8-L11, and a target cell can be assigned to a fourth bin if its single adjacent cell is determined to have a state from L12-L15.

As yet another example, in a 1-bit 2-sided 2BCR implementation in which $B=2$, the number of bins is 4 (i.e., 2^2) and each bin defines a respective cell state information condition. Illustratively, for the QLC cell implementation, a target cell can be assigned to a first bin if the adjacent cell connected to adjacent wordline WL_{n-1} and the adjacent cell connected to adjacent wordline WL_{n+1} are each determined to have a state from L0-L7. A target cell can be assigned to a second bin if the adjacent cell connected to adjacent wordline WL_{n-1} is determined to have a state from L0-L7 and the adjacent cell connected to adjacent wordline WL_{n+1} is determined to have a state from L8-L15. A target cell can

be assigned to a third bin if the adjacent cell connected to adjacent wordline WL_{n-1} is determined to have a state from L8-L15 and the adjacent cell connected to adjacent wordline WL_{n+1} is determined to have a state from L0-L7. A target cell can be assigned to a fourth bin if the adjacent cell connected to adjacent wordline WL_{n-1} and the adjacent cell connected to adjacent wordline WL_{n+1} are each determined to have a state from L8-L15.

As yet another example, in a 4BCR implementation in which $B=4$, the number of bins is 16 (i.e., 2^4) and each bin defines a respective cell state information condition. Illustratively, for the QLC cell implementation, a target cell can be assigned to a first bin if the adjacent cell connected to adjacent wordline WL_{n-1} and the adjacent cell connected to adjacent wordline WL_{n+1} each have a state from L0-L3. A target cell can be assigned to a second bin if the adjacent cell connected to adjacent wordline WL_{n-1} is determined to have a state from L0-L3 and the adjacent cell connected to adjacent wordline WL_{n+1} is determined to have a state from L4-L7. A target cell can be assigned to a third bin if the adjacent cell connected to adjacent wordline WL_{n-1} is determined to have a state L0-L3 and the adjacent cell connected to adjacent wordline WL_{n+1} is determined to have a state from L8-L11. A target cell can be assigned to a fourth bin if the adjacent cell connected to adjacent wordline WL_{n-1} is determined to have a state from L0-L3 and the adjacent cell connected to adjacent wordline WL_{n+1} is determined to have a state from L12-L15. The remaining 12 bins can be generated with other similar combinations of states.

Each bin is assigned a respective read level offset for reading the target cells assigned to the bin. Each read level offset accounts for the effect that the state of each cell of the group of adjacent cells has on the respective target cell. Typically, the read level offset assigned to a bin is empirically determined through observation or experimentation of the effect that adjacent cell(s) with varying states have on respective target cells. The bin assignments and corresponding read level offsets can be stored in a metadata area of the device. Accordingly, the bins and read level offset assignments can be pre-determined before assigning target cells to bins.

The controller can then cause the target cells to be read using respective ones of the read level offsets. For example, the controller can cause target cells of a first bin, which form a first target cell V_T sub-distribution, to be read at a first read level. The first read level is determined by locating a center read level, and applying the read level offset assigned to the first bin to the center read level. The center read level can have a voltage magnitude located within the valley between a pair of target cell V_T distributions, and the read level offset can adjust the center read level such that the first read level can have a voltage magnitude located in the valley to the left of the first target cell V_T sub-distribution. The other target cells can be read similarly.

In view of the above, the total number of reads performed can be equal to the number of bins or target cell V_T sub-distributions. For example, in a 1BCR implementation, the target cells of the first bin can be read using the first read level offset and the target cells of the second bin can be read using the second read level offset as described above, for a total of 2 reads. As another example, in a 2BCR implementation (e.g., 1-bit 2-sided or 2-bit 1-sided), the target cells in each of the 4 bins can be read using a respective read level offset, for a total of 4 reads. As yet another example, in a 4BCR implementation, the target cells in each of the 16 bins can be read using a respective read level offset, for a total of 16 reads.

As described above, 4BCR can be implemented by reading 16 bins of target cells each with a respective read level offset. During each of the 16 reads, a number of page reads are performed. For example, to read a QLC target cell, LP could be read 4 times, UP could be read 4 times, TP could be read 3 times, or XP can be read 4 times. This results in an average of 3.75 reads per page. Therefore, the total number of reads that are performed during QLC 4BCR can be about 66 reads (16 read operations multiplied by 3.75 average reads per page on the target cell, in addition to the 3 reads performed on each of the adjacent cells). The potentially large number of reads that are performed during 4BCR can introduce a performance penalty due to the amount of time needed to perform the 16 reads, and can impact memory device reliability due to phenomena such as read disturb. These negative impacts of corrective read can be amplified for higher bit corrective read implementations, which can require even more target cell reads. Accordingly, reducing the number of reads performed during corrective read can improve memory device performance and reliability.

It may be observed that pairs of adjacent cells can have similar retention loss with respect to a target cell. For example, assume that a target cell connected to a target wordline WL_n is a QLC cell. A pair of adjacent cells including an L0 cell connected to adjacent wordline WL_{n-1} and an L15 cell connected to adjacent wordline WL_{n+1} , a pair of adjacent cells including an L15 cell connected to adjacent wordline WL_{n-1} and an L0 cell connected to adjacent wordline WL_{n+1} , and a pair of adjacent cells including L7 cells can each result in similar retention loss. Accordingly, it may be inefficient to assign each target cell to a respective bin having a cell state information condition defined by separately evaluating the cell state information for the cells of the respective pair of adjacent wordline cells on an individual or isolated basis.

Aspects of the present disclosure address the above and other deficiencies by performing corrective reads with improved recovery from data retention loss. Embodiments described herein can, during a corrective read operation, assign a target cell to a particular state information bin ("bin") by applying a pre-defined operation) to the cell state information obtained from each cell of a pair of adjacent cells. The bin is assigned with a particular read level offset for reading the target cell during the read operation.

In some embodiments, the cell state information of a first cell connected to the adjacent wordline WL_{n-1} is a first threshold voltage V_{T1} indicative of a state of the first cell, and the cell state information for a second cell connected to the adjacent wordline WL_{n+1} is a second threshold voltage V_{T2} indicative of a state of the second cell. The pre-defined operation can be a linear combination of V_{T1} and V_{T2} . In some embodiments, the linear combination is the sum $V_{T1} + V_{T2}$. In some embodiments, the linear combination is a weighted sum $xV_{T1} + yV_{T2}$, where $x \neq y$. The weighted sum can be used to reflect more source side WL or drain side WL weight (e.g., $x=1.5$ and $y=1$). The pre-defined operation is not limited to linear combinations. For example, the pre-defined operation can be extended to include other mathematical operations or functions. Such mathematical operations can utilize read level of each target cell in conjunction with the V_{T1} and V_{T2} of the respective pair of adjacent cells. Further details performing corrective reads implementing efficient binning are described herein below with reference to FIGS. 1-6.

Advantages of the present disclosure include, but are not limited to, improved memory device performance and reli-

ability. For example, by applying the pre-defined operation during target cell bin assignment, the target cells can be assigned to a reduced number of bins. The bin reduction can lead to a reduction of the total number of reads performed as compared to typical corrective read, which can reduce read disturb effects. Moreover, the amount of time that is saved by reducing the number of reads can be invested into additional adjacent cell reads to improve corrective read effectiveness. Improved corrective read effectiveness can lead to improved RWB.

FIG. 1A illustrates an example computing system **100** that includes a memory sub-system **110** in accordance with some embodiments of the present disclosure. The memory sub-system **110** can include media, such as one or more volatile memory devices (e.g., memory device **140**), one or more non-volatile memory devices (e.g., memory device **130**), or a combination of such.

A memory sub-system **110** can be a storage device, a memory module, or a combination of a storage device and memory module. Examples of a storage device include a solid-state drive (SSD), a flash drive, a universal serial bus (USB) flash drive, an embedded Multi-Media Controller (eMMC) drive, a Universal Flash Storage (UFS) drive, a secure digital (SD) card, and a hard disk drive (HDD). Examples of memory modules include a dual in-line memory module (DIMM), a small outline DIMM (SO-DIMM), and various types of non-volatile dual in-line memory modules (NVDIMMs).

The computing system **100** can be a computing device such as a desktop computer, laptop computer, network server, mobile device, a vehicle (e.g., airplane, drone, train, automobile, or other conveyance), Internet of Things (IoT) enabled device, embedded computer (e.g., one included in a vehicle, industrial equipment, or a networked commercial device), or such computing device that includes memory and a processing device.

The computing system **100** can include a host system **120** that is coupled to one or more memory sub-systems **110**. In some embodiments, the host system **120** is coupled to multiple memory sub-systems **110** of different types. FIG. 1A illustrates one example of a host system **120** coupled to one memory sub-system **110**. As used herein, “coupled to” or “coupled with” generally refers to a connection between components, which can be an indirect communicative connection or direct communicative connection (e.g., without intervening components), whether wired or wireless, including connections such as electrical, optical, magnetic, etc.

The host system **120** can include a processor chipset and a software stack executed by the processor chipset. The processor chipset can include one or more cores, one or more caches, a memory controller (e.g., NVDIMM controller), and a storage protocol controller (e.g., PCIe controller, SATA controller). The host system **120** uses the memory sub-system **110**, for example, to write data to the memory sub-system **110** and read data from the memory sub-system **110**.

The host system **120** can be coupled to the memory sub-system **110** via a physical host interface. Examples of a physical host interface include, but are not limited to, a serial advanced technology attachment (SATA) interface, a peripheral component interconnect express (PCIe) interface, universal serial bus (USB) interface, Fibre Channel, Serial Attached SCSI (SAS), a double data rate (DDR) memory bus, Small Computer System Interface (SCSI), a dual in-line memory module (DIMM) interface (e.g., DIMM socket interface that supports Double Data Rate (DDR)), etc. The physical host interface can be used to transmit data between

the host system **120** and the memory sub-system **110**. The host system **120** can further utilize an NVM Express (NVMe) interface to access components (e.g., memory devices **130**) when the memory sub-system **110** is coupled with the host system **120** by the physical host interface (e.g., PCIe bus). The physical host interface can provide an interface for passing control, address, data, and other signals between the memory sub-system **110** and the host system **120**. FIG. 1A illustrates a memory sub-system **110** as an example. In general, the host system **120** can access multiple memory sub-systems via a same communication connection, multiple separate communication connections, and/or a combination of communication connections.

The memory devices **130**, **140** can include any combination of the different types of non-volatile memory devices and/or volatile memory devices. The volatile memory devices (e.g., memory device **140**) can be, but are not limited to, random access memory (RAM), such as dynamic random access memory (DRAM) and synchronous dynamic random access memory (SDRAM).

Some examples of non-volatile memory devices (e.g., memory device **130**) include a negative-and (NAND) type flash memory and write-in-place memory, such as a three-dimensional cross-point (“3D cross-point”) memory device, which is a cross-point array of non-volatile memory cells. A cross-point array of non-volatile memory cells can perform bit storage based on a change of bulk resistance, in conjunction with a stackable cross-gridded data access array. Additionally, in contrast to many flash-based memories, cross-point non-volatile memory can perform a write in-place operation, where a non-volatile memory cell can be programmed without the non-volatile memory cell being previously erased. NAND type flash memory includes, for example, two-dimensional NAND (2D NAND) and three-dimensional NAND (3D NAND).

Each of the memory devices **130** can include one or more arrays of memory cells. One type of memory cell, for example, single level memory cells (SLC) can store one bit per memory cell. Other types of memory cells, such as multi-level memory cells (MLCs), triple level memory cells (TLCs), quad-level memory cells (QLCs), and penta-level memory cells (PLCs) can store multiple bits per memory cell. In some embodiments, each of the memory devices **130** can include one or more arrays of memory cells such as SLCs, MLCs, TLCs, QLCs, PLCs or any combination of such. In some embodiments, a particular memory device can include an SLC portion, and an MLC portion, a TLC portion, a QLC portion, or a PLC portion of memory cells. The memory cells of the memory devices **130** can be grouped as pages that can refer to a logical unit of the memory device used to store data. With some types of memory (e.g., NAND), pages can be grouped to form blocks.

Although non-volatile memory components such as a 3D cross-point array of non-volatile memory cells and NAND type flash memory (e.g., 2D NAND, 3D NAND) are described, the memory device **130** can be based on any other type of non-volatile memory, such as read-only memory (ROM), phase change memory (PCM), self-selecting memory, other chalcogenide based memories, ferroelectric transistor random-access memory (FeTRAM), ferroelectric random access memory (FeRAM), magneto random access memory (MRAM), Spin Transfer Torque (STT)-MRAM, conductive bridging RAM (CBRAM), resistive random access memory (RRAM), oxide based RRAM (OxRAM), negative-or (NOR) flash memory, or electrically erasable programmable read-only memory (EEPROM).

11

A memory sub-system controller **115** (or controller **115** for simplicity) can communicate with the memory devices **130** to perform operations such as reading data, writing data, or erasing data at the memory devices **130** and other such operations. The memory sub-system controller **115** can include hardware such as one or more integrated circuits and/or discrete components, a buffer memory, or a combination thereof. The hardware can include a digital circuitry with dedicated (i.e., hard-coded) logic to perform the operations described herein. The memory sub-system controller **115** can be a microcontroller, special purpose logic circuitry (e.g., a field programmable gate array (FPGA), an application specific integrated circuit (ASIC), etc.), or other suitable processor.

The memory sub-system controller **115** can include a processing device, which includes one or more processors (e.g., processor **117**), configured to execute instructions stored in a local memory **119**. In the illustrated example, the local memory **119** of the memory sub-system controller **115** includes an embedded memory configured to store instructions for performing various processes, operations, logic flows, and routines that control operation of the memory sub-system **110**, including handling communications between the memory sub-system **110** and the host system **120**.

In some embodiments, the local memory **119** can include memory page buffers storing memory pointers, fetched data, etc. The local memory **119** can also include read-only memory (ROM) for storing micro-code. While the example memory sub-system **110** in FIG. 1 has been illustrated as including the memory sub-system controller **115**, in another embodiment of the present disclosure, a memory sub-system **110** does not include a memory sub-system controller **115**, and can instead rely upon external control (e.g., provided by an external host, or by a processor or controller separate from the memory sub-system).

In general, the memory sub-system controller **115** can receive commands or operations from the host system **120** and can convert the commands or operations into instructions or appropriate commands to achieve the desired access to the memory devices **130**. The memory sub-system controller **115** can be responsible for other operations such as wear leveling operations, garbage collection operations, error detection and error-correcting code (ECC) operations, encryption operations, caching operations, and address translations between a logical address (e.g., a logical block address (LBA), namespace) and a physical address (e.g., physical block address) that are associated with the memory devices **130**. The memory sub-system controller **115** can further include host interface circuitry to communicate with the host system **120** via the physical host interface. The host interface circuitry can convert the commands received from the host system into command instructions to access the memory devices **130** as well as convert responses associated with the memory devices **130** into information for the host system **120**.

The memory sub-system **110** can also include additional circuitry or components that are not illustrated. In some embodiments, the memory sub-system **110** can include a cache or buffer (e.g., DRAM) and address circuitry (e.g., a row decoder and a column decoder) that can receive an address from the memory sub-system controller **115** and decode the address to access the memory devices **130**.

In some embodiments, the memory devices **130** include local media controllers **135** that operate in conjunction with memory sub-system controller **115** to execute operations on one or more memory cells of the memory devices **130**. An

12

external controller (e.g., memory sub-system controller **115**) can externally manage the memory device **130** (e.g., perform media management operations on the memory device **130**). In some embodiments, memory sub-system **110** is a managed memory device, which is a raw memory device **130** having control logic (e.g., local controller **132**) on the die and a controller (e.g., memory sub-system controller **115**) for media management within the same memory device package. An example of a managed memory device is a managed NAND (MNAND) device.

The local media controller **135** can implement a corrective read (CR) component **137** that can perform corrective reads implementing efficient binning to improve recovery from data retention loss. For example, the CR component **137** can initiate a corrective read operation to read a set of target cells connected to a target wordline. For example, the CR component **137** can initiate the corrective read operation in response to receiving a request (e.g., read command) via the memory sub-system controller **115** to read the set of target cells.

The CR component **137** can obtain, for each target cell, cell state information from each cell of respective pair of adjacent cells. Each cell of a pair of adjacent cells neighbors a respective target cell. For example, if the memory device **130** is a 3D memory device, one cell of a pair of adjacent cells can be located directly above its target cell and another cell of the pair of adjacent cells can be located directly below its target cell. Each cell of the pair of adjacent cells is connected to a respective wordline of a pair of adjacent wordlines neighboring the target wordline. For example, if the memory device **130** is a 3D memory device, one wordline of the pair of adjacent wordlines can be directly above the target wordline and the other wordline of the pair of adjacent wordlines can be located directly below the target wordline.

In some embodiments, the cell state information for each cell of the pair of adjacent cells includes 2-bit cell state information, such that the cell state information for the pair of adjacent cells includes 4-bit cell state information. The cell state information for each pair of adjacent cells can be stored in a respective page buffer. Each page buffer can be connected to its respective pair of adjacent cells and respective target cell via a respective bitline. In some embodiments, the cell state information for a first cell of the pair of adjacent cells (e.g., connected to the adjacent wordline WL_{n-1}) is a first V_T (V_{T1}) indicative of a state of the first cell, and the cell state information of a second cell of the pair of adjacent cells (e.g., connected to the adjacent wordline WL_{n+1}) is a second V_T (V_{T2}) indicative of a state of the second cell.

The CR component **137** can then assign each target cell to a respective state information bin ("bin") by applying a pre-defined operation to the cell state information obtained from each cell of the respective pair of adjacent cells. Each bin defines a respective cell state information condition, and a target cell is assigned to a bin if the cell(s) within its respective pair of adjacent cells satisfy the cell state information condition for the bin. Thus, the bins divide each target cell V_T distribution into a number of respective target cell V_T sub-distributions, where each target cell V_T sub-distribution is formed from the target cells assigned to a respective bin.

In some embodiments, the pre-defined operation is a linear combination of V_{T1} and V_{T2} . For example, the linear combination can be the sum $V_{T1} + V_{T2}$. As another example, the linear combination can be a weighted sum $xV_{T1} + yV_{T2}$, where $x \neq y$. The weighted sum can be used to reflect more

13

source side WL or drain side WL weight (e.g., $x=1.5$ and $y=1$). The pre-defined operation is not limited to linear combinations. For example, the pre-defined operation can be extended to include other mathematical operations or functions. Such mathematical operations can utilize read level of each target cell in conjunction with the V_{T1} and V_{T2} of the respective pair of adjacent cells.

The CR component 137 can then cause each target cell of the set of target cells to be read using respective ones of the read level offsets. For example, the controller can cause target cells of a first bin, which form a first target cell V_T sub-distribution, to be read at a first read level. The first read level is determined by locating a center read level, and applying the read level offset assigned to the first bin to the center read level. The center read level can have a voltage magnitude located within the valley between a pair of target cell V_T distributions, and the read level offset can adjust the center read level such that the first read level can have a voltage magnitude located in the valley to the left of the first target cell V_T sub-distribution. The other target cells can be read similarly.

By applying the pre-defined operation during target cell bin assignment, the target cells can be assigned to a reduced number of bins. The bin reduction can lead to a reduction of the total number of reads performed as compared to typical corrective read, which can reduce read disturb effects. Moreover, the amount of time that is saved by reducing the number of read can be invested into additional adjacent cell reads to improve corrective read effectiveness.

For example, in a typical QLC implementation of 4BCR, there may be 16 total bins, and the total average number of reads per page can be about 66. As will be described in further detail below with reference to FIGS. 6A-6D, if the number of reads performed with respect to each adjacent cell to obtain the cell state information is 3, then the number of bins can be reduced from 16 to 7. If the number of reads performed with respect to each adjacent cell to obtain the cell state information is 4, then the number of bins can be reduced from 16 to 9. If the number of reads performed with respect to each adjacent cell to obtain the cell state information is 5, then the number of bins can be reduced from 16 to 11. If the number of reads performed with respect to each adjacent cell to obtain the cell state information is 6, then the number of bins can be reduced from 16 to 13. Each of these reduced bin examples can result in fewer than 66 average number of reads per page. Further details regarding the operations of the CR component 137 will be described below with reference to FIGS. 4-6D.

FIG. 1B is a simplified block diagram of a first apparatus, in the form of a memory device 130, in communication with a second apparatus, in the form of a memory sub-system controller 115 of a memory sub-system (e.g., memory sub-system 110 of FIG. 1), according to an embodiment. Some examples of electronic systems include personal computers, personal digital assistants (PDAs), digital cameras, digital media players, digital recorders, games, appliances, vehicles, wireless devices, mobile telephones and the like. The memory sub-system controller 115 (e.g., a controller external to the memory device 130), may be a memory controller or other external host device.

Memory device 130 includes an array of memory cells 104 logically arranged in rows and columns. Memory cells of a logical row are connected to the same access line (e.g., a wordline) while memory cells of a logical column are selectively connected to the same data line (e.g., a bitline). A single access line may be associated with more than one logical row of memory cells and a single data line may be

14

associated with more than one logical column. Memory cells (not shown in FIG. 1B) of at least a portion of array of memory cells 104 are capable of being programmed to one of at least two target data states.

Row decode circuitry 108 and column decode circuitry 112 are provided to decode address signals. Address signals are received and decoded to access the array of memory cells 104. Memory device 130 also includes input/output (I/O) control circuitry 160 to manage input of commands, addresses and data to the memory device 130 as well as output of data and status information from the memory device 130. An address page buffer 114 is in communication with I/O control circuitry 160 and row decode circuitry 108 and column decode circuitry 112 to latch the address signals prior to decoding. A command page buffer 124 is in communication with I/O control circuitry 160 and local media controller 135 to latch incoming commands.

A controller (e.g., the local media controller 135 internal to the memory device 130) controls access to the array of memory cells 104 in response to the commands and generates status information for the external memory sub-system controller 115, i.e., the local media controller 135 is configured to perform access operations (e.g., read operations, programming operations and/or erase operations) on the array of memory cells 104. The local media controller 135 is in communication with row decode circuitry 108 and column decode circuitry 112 to control the row decode circuitry 108 and column decode circuitry 112 in response to the addresses. In one embodiment, local media controller 135 includes the CR component 137, which can implement the defect detection described herein during an erase operation on memory device 130.

The local media controller 135 is also in communication with a cache page buffer 118. Cache page buffer 118 latches data, either incoming or outgoing, as directed by the local media controller 135 to temporarily store data while the array of memory cells 104 is busy writing or reading, respectively, other data. During a program operation (e.g., write operation), data may be passed from the cache page buffer 118 to the data page buffer 170 for transfer to the array of memory cells 104; then new data may be latched in the cache page buffer 118 from the I/O control circuitry 160. During a read operation, data may be passed from the cache page buffer 118 to the I/O control circuitry 160 for output to the memory sub-system controller 115; then new data may be passed from the data page buffer 170 to the cache page buffer 118. The cache page buffer 118 and/or the data page buffer 170 may form (e.g., may form a portion of) a page buffer of the memory device 130. A page buffer may further include sensing devices (not shown in FIG. 1B) to sense a data state of a memory cell of the array of memory cells 204, e.g., by sensing a state of a data line connected to that memory cell. A status page buffer 122 may be in communication with I/O control circuitry 160 and the local memory controller 135 to latch the status information for output to the memory sub-system controller 115.

Memory device 130 receives control signals at the memory sub-system controller 115 from the local media controller 135 over a control link 132. For example, the control signals can include a chip enable signal CE #, a command latch enable signal CLE, an address latch enable signal ALE, a write enable signal WE #, a read enable signal RE #, and a write protect signal WP #. Additional or alternative control signals (not shown) may be further received over control link 132 depending upon the nature of the memory device 130. In one embodiment, memory device 130 receives command signals (which represent com-

15

mands), address signals (which represent addresses), and data signals (which represent data) from the memory sub-system controller **115** over a multiplexed input/output (I/O) bus **136** and outputs data to the memory sub-system controller **115** over I/O bus **136**.

For example, the commands may be received over input/output (I/O) pins [7:0] of I/O bus **136** at I/O control circuitry **160** and may then be written into command page buffer **124**. The addresses may be received over input/output (I/O) pins [7:0] of I/O bus **136** at I/O control circuitry **160** and may then be written into address page buffer **114**. The data may be received over input/output (I/O) pins [7:0] for an 8-bit device or input/output (I/O) pins [15:0] for a 16-bit device at I/O control circuitry **160** and then may be written into cache page buffer **118**. The data may be subsequently written into data page buffer **170** for programming the array of memory cells **104**.

In an embodiment, cache page buffer **118** may be omitted, and the data may be written directly into data page buffer **170**. Data may also be output over input/output (I/O) pins [7:0] for an 8-bit device or input/output (I/O) pins [15:0] for a 16-bit device. Although reference may be made to I/O pins, they may include any conductive node providing for electrical connection to the memory device **130** by an external device (e.g., the memory sub-system controller **115**), such as conductive pads or conductive bumps as are commonly used.

It will be appreciated by those skilled in the art that additional circuitry and signals can be provided, and that the memory device **130** of FIGS. 1A-1B has been simplified. It should be recognized that the functionality of the various block components described with reference to FIGS. 1A-1B may not necessarily be segregated to distinct components or component portions of an integrated circuit device. For example, a single component or component portion of an integrated circuit device could be adapted to perform the functionality of more than one block component of FIGS. 1A-1B. Alternatively, one or more components or component portions of an integrated circuit device could be combined to perform the functionality of a single block component of FIGS. 1A-1B. Additionally, while specific I/O pins are described in accordance with popular conventions for receipt and output of the various signals, it is noted that other combinations or numbers of I/O pins (or other I/O node structures) may be used in the various embodiments.

FIGS. 2A-2C are diagrams of portions of an example array of memory cells included in a memory device, in accordance with some embodiments of the present disclosure. For example, FIG. 2A is a schematic of a portion of an array of memory cells **200A** as could be used in a memory device (e.g., as a portion of array of memory cells **104**). Memory array **200A** includes access lines, such as wordlines **202₀** to **202_N**, and a data line, such as bitline **204**. The wordlines **202** may be connected to global access lines (e.g., global wordlines), not shown in FIG. 2A, in a many-to-one relationship. For some embodiments, memory array **200A** may be formed over a semiconductor that, for example, may be conductively doped to have a conductivity type, such as a p-type conductivity, e.g., to form a p-well, or an n-type conductivity, e.g., to form an n-well.

Memory array **200A** can be arranged in rows each corresponding to a respective wordline **202** and columns each corresponding to a respective bitline **204**. Rows of memory cells **208** can be divided into one or more groups of physical pages of memory cells **208**, and physical pages of memory cells **208** can include every other memory cell **208** commonly connected to a given wordline **202**. For example,

16

memory cells **208** commonly connected to wordline **202_N** and selectively connected to even bitlines **204** (e.g., bitlines **204₀**, **204₂**, **204₄**, etc.) may be one physical page of memory cells **208** (e.g., even memory cells) while memory cells **208** commonly connected to wordline **202_N** and selectively connected to odd bitlines **204** (e.g., bitlines **204₁**, **204₃**, **204₅**, etc.) may be another physical page of memory cells **208** (e.g., odd memory cells). Although bitlines **204₃**-**204₅** are not explicitly depicted in FIG. 2A, it is apparent from the figure that the bitlines **204** of the array of memory cells **200A** may be numbered consecutively from bitline **204₀** to bitline **204_M**. Other groupings of memory cells **208** commonly connected to a given wordline **202** may also define a physical page of memory cells **208**. For certain memory devices, all memory cells commonly connected to a given wordline might be deemed a physical page of memory cells. The portion of a physical page of memory cells (which, in some embodiments, could still be the entire row) that is read during a single read operation or programmed during a single programming operation (e.g., an upper or lower page of memory cells) might be deemed a logical page of memory cells. A block of memory cells may include those memory cells that are configured to be erased together, such as all memory cells connected to wordlines **202₀**-**202_N** (e.g., all strings **206** sharing common wordlines **202**). Unless expressly distinguished, a reference to a page of memory cells herein refers to the memory cells of a logical page of memory cells.

Each column can include a string of series-connected memory cells (e.g., non-volatile memory cells), such as one of strings **206₀** to **206_M**. Each string **206** can be connected (e.g., selectively connected) to a source line **216** (SRC) and can include memory cells **208₀** to **208_N**. The memory cells **208** of each string **206** can be connected in series between a select gate **210**, such as one of the select gates **210₀** to **210_M**, and a select gate **212**, such as one of the select gates **212₀** to **212_M**. In some embodiments, the select gates **210₀** to **210_M** are source-side select gates (SGS) and the select gates **212₀** to **212_M** are drain-side select gates. Select gates **210₀** to **210_M** can be connected to a select line **214** (e.g., source-side select line) and select gates **212₀** to **212_M** can be connected to a select line **215** (e.g., drain-side select line). The select gates **210** and **212** might represent a plurality of select gates connected in series, with each select gate in series configured to receive a same or independent control signal. A source of each select gate **210** can be connected to SRC **216**, and a drain of each select gate **210** can be connected to a memory cell **208₀** of the corresponding string **206**. Therefore, each select gate **210** can be configured to selectively connect a corresponding string **206** to SRC **216**. A control gate of each select gate **210** can be connected to select line **214**. The drain of each select gate **212** can be connected to the bitline **204** for the corresponding string **206**. The source of each select gate **212** can be connected to a memory cell **208_N** of the corresponding string **206**. Therefore, each select gate **212** might be configured to selectively connect a corresponding string **206** to the bitline **204**. A control gate of each select gate **212** can be connected to select line **215**.

In some embodiments, and as will be described in further detail below with reference to FIG. 2B, the memory array in FIG. 2A is a three-dimensional memory array, in which the strings **206** extend substantially perpendicular to a plane containing SRC **216** and to a plane containing a plurality of bitlines **204** that can be substantially parallel to the plane containing SRC **216**.

17

FIG. 2B is another schematic of a portion of an array of memory cells 200B (e.g., a portion of the array of memory cells 104) arranged in a three-dimensional memory array structure. The three-dimensional memory array 200B may incorporate vertical structures which may include semiconductor pillars where a portion of a pillar may act as a channel region of the memory cells of strings 206. The strings 206 may be each selectively connected to a bit line 204₀-204_M by a select gate 212 and to the SRC 216 by a select gate 210. Multiple strings 206 can be selectively connected to the same bitline 204. Subsets of strings 206 can be connected to their respective bitlines 204 by biasing the select lines 215₀-215_L to selectively activate particular select gates 212 each between a string 206 and a bitline 204. The select gates 210 can be activated by biasing the select line 214. Each wordline 202 may be connected to multiple rows of memory cells of the memory array 200B. Rows of memory cells that are commonly connected to each other by a particular wordline 202 may collectively be referred to as tiers.

FIG. 2C is a diagram of a portion of an array of memory cells 200C (e.g., a portion of the array of memory cells 104). Channel regions (e.g., semiconductor pillars) 238₀₀ and 238₀₁ represent the channel regions of different strings of series-connected memory cells (e.g., strings 206 of FIGS. 2A-2B) selectively connected to the bitline 204₀. Similarly, channel regions 238₁₀ and 238₁₁ represent the channel regions of different strings of series-connected memory cells (e.g., NAND strings 206 of FIGS. 2A-2B) selectively connected to the bitline 204₁. A memory cell (not depicted in FIG. 2C) may be formed at each intersection of an wordline 202 and a channel region 238, and the memory cells corresponding to a single channel region 238 may collectively form a string of series-connected memory cells (e.g., a string 206 of FIGS. 2A-2B). Additional features might be common in such structures, such as dummy wordlines, segmented channel regions with interposed conductive regions, etc.

FIG. 3 is a diagram of an example three-dimensional (3D) replacement gate memory device ("device") 300, in accordance with some embodiments of the present disclosure. However, the embodiments described herein can be applied to any suitable memory device. As shown, the device 300 includes a bitline 310, pillars 320-1 and 320-2, select gates (SGs) 330-1 and 330-2, a source line (SRC) 340, and WL groups 350-1, 350-2, 360-1 and 360-2. More specifically, WL groups 350-1 and 350-2 are dummy WL groups, and WL groups 360-1 and 360-2 are active WL groups. WL group 350-1 includes dummy WLs 352-1 through 366-1, WL group 350-2 includes dummy WLs 352-2 through 356-2, WL group 360-1 includes active WLs 362-1 and 364-1, and WL group 360-2 includes active WLs 362-2, 364-2 and 366-2. However, such an example should not be considered limiting. A dummy WL corresponds to memory cells that do not store data and are included to satisfy processing margins, while an active WL corresponds to memory cells that store data.

As further shown, a WL 370 is provided. In some embodiments, the device 300 is a multiple deck device, in which WL groups 350-1 and 360-1 are associated with a first deck (e.g., an upper deck) of the device 300 and the WL groups 350-2 and 360-2 are associated with a second deck (e.g., a lower deck) of the device 300, such that the WL 370 corresponds to a dummy WL separating the WL groups 360-1 and 360-2. In other embodiments, the device 300 is a "single deck" device, in which the WL groups 360-1 and 360-2 are not arranged in decks. Here, the WL 370 can be an active WL within one of the WL groups 360-1 or 360-2.

18

FIG. 4 is a diagram of a portion of a memory array 400, in accordance with some embodiments. The memory array 400 can include any suitable number of wordlines (WLs). For example, as shown, the memory array 400 includes a number of wordlines WL 410-1 through WL 410-(N+2). Each of the WLs 410-1 through 410-(N+2) is connected to a respective set of cells. Each of the WLs 410-1 through 410-(N+2) is adjacent to at least one WL. For example, WL 410-(N+1) and WL 410-(N-1) are each adjacent wordlines with respect to WL 410-N. The memory array 400 further includes select gate (SG) 420-1 and SG 420-2. In some embodiments, SG 420-1 is a source-side SG (SGS) and SG 420-2 is a drain-side SG (SGD).

The memory array 400 further includes a number of bitlines (BLs) including BL 430-1 through 430-4 and a number of page buffers including page buffers 440-1 through 440-4. Each of the page buffers is connected to a respective one of the bitlines. Although only 4 bitlines 430-1 through 410-3 and page buffers 440-1 through 440-4 are shown, the memory array 400 can include any suitable number of bitlines and page buffers.

In this illustrative example, a set of target cells 450 is selected to be read. The set of target cells 450 includes a number of cells of the target wordline WL 410-N. Each target cell of the set of target cells 450 is adjacent to a pair of adjacent cells. More specifically, the pair of adjacent cells for a particular target cell includes the cell connected to WL 410-(N+1) that is directly above the target cell, and the cell connected to WL 410-(N-1) that is directly below the target cell. That is, a target cell of the set of target cells 450 is connected to a same one of the bitlines as its respective pair of adjacent cells.

A local media controller (e.g., local media controller 135 of FIGS. 1A-1B) can perform corrective read with respect to the set of target cells 450. To perform the corrective read with respect to the set of target cells 450, the local media controller can, for each target cell of the set of target cells 450, cause cell state information to be obtained for each cell of the respective pair of adjacent cells. The cell state information for each cell can include a V_T value indicative of the state of the cell (e.g., program state or erase state). For each pair of adjacent cells, the cell state information can include a first V_T value (V_{T1}) indicative of a state of a first cell of the pair of adjacent cells, and a second V_T value (V_{T2}) indicative of a state of a second cell of the pair of adjacent cells.

The cell state information for each pair of adjacent cells can be maintained (e.g. stored) in a respective page buffer. For example, the page buffer 440-1 can maintain the cell state information of the cells connected to WL 410-(N-1) and WL 410-(N+1) that are connected to BL 430-1. The page buffer 440-2 can maintain the cell state information of the cells connected to WL 410-(N-1) and WL 410-(N+1) that are connected to BL 430-2. The page buffer 440-3 can maintain the cell state information of the cells connected to WL 410-(N-1) and WL 410-(N+1) that are connected to BL 430-3. The page buffer 440-4 can maintain the cell state information of the cells connected to WL 410-(N-1) and WL 410-(N+1) that are connected to BL 430-4.

The local media controller can then assign each target cell of the set of target cells 450 to a respective state information bin ("bin") by applying a pre-defined operation to the cell state information V_{T1} and V_{T2} maintained in each of the page buffers 440-1 through 440-4. Each bin defines a respective cell state information condition, and a target cell is assigned to a bin if the cell(s) within its respective pair of adjacent cells satisfy the cell state information condition for the bin.

Thus, the bins divide each target cell V_T distribution into a number of respective target cell V_T sub-distributions, where each target cell V_T sub-distribution is formed from the target cells assigned to a respective bin.

In some embodiments, the pre-defined operation is a linear combination of V_{T1} and V_{T2} . For example, the linear combination can be the sum $V_{T1} + V_{T2}$. As another example, the linear combination can be a weighted sum $xV_{T1} + yV_{T2}$, where $x \neq y$. The weighted sum can be used to reflect more source side WL or drain side WL weight (e.g., $x=1.5$ and $y=1$). The pre-defined operation is not limited to linear combinations. For example, the pre-defined operation can be extended to include other mathematical operations or functions. Such mathematical operations can utilize read level of each target cell of the set of target cells **450** in conjunction with the V_{T1} and V_{T2} of the respective pair of adjacent cells.

The controller can then cause each target cell of the set of target cells **450** to be read using respective ones of the read level offsets. For example, the controller can cause target cells of a first bin, which form a first target cell V_T sub-distribution, to be read at a first read level. The first read level is determined by locating a center read level, and applying the read level offset assigned to the first bin to the center read level. The center read level can have a voltage magnitude located within the valley between a pair of target cell V_T distributions, and the read level offset can adjust the center read level such that the first read level can have a voltage magnitude located in the valley to the left of the first target cell V_T sub-distribution. The other target cells can be read similarly.

By applying the pre-defined operation during target cell bin assignment, the target cells can be assigned to a reduced number of bins. The bin reduction can lead to a reduction of the total number of reads performed as compared to typical corrective read, which can reduce read disturb effects. Moreover, the amount of time that is saved by reducing the number of reads can be invested into additional adjacent cell reads to improve corrective read effectiveness. Further details regarding performing corrective reads with improved recovery from data retention loss be described in further detail below with reference to FIGS. **5-6D**.

FIG. **5** is a flow diagram of an example method **500** to perform a corrective read with improved recovery from data retention loss, in accordance with some embodiments of the present disclosure. The method **500** can be performed by processing logic that can include hardware (e.g., processing device, circuitry, dedicated logic, programmable logic, microcode, hardware of a device, integrated circuit, etc.), software (e.g., instructions run or executed on a processing device), or a combination thereof. In some embodiments, the method **500** is performed by the CR component **137** of FIGS. **1A-1B**. Although shown in a particular sequence or order, unless otherwise specified, the order of the processes can be modified. Thus, the illustrated embodiments should be understood only as examples, and the illustrated processes can be performed in a different order, and some processes can be performed in parallel. Additionally, one or more processes can be omitted in various embodiments. Thus, not all processes are required in every embodiment. Other process flows are possible.

At operation **510**, a read operation is initiated with respect to a set of target cells. For example, processing logic can cause the corrective read to be initiated with respect to the set of target cells in response to receiving a request to perform corrective read. The request can be a read command received via a memory sub-system controller (e.g., the

memory sub-system controller **115** of FIGS. **1A-1B**). Each target cell of the set of target cells is connected to a target wordline.

At operation **520**, cell state information is obtained. For example, the processing logic can cause cell state information to be obtained from respective pairs of adjacent cells. Each pair of adjacent cells includes a first cell and a second cell adjacent to a respective target cell of the set of target cells. The first cell is connected to a first wordline adjacent to the target wordline and the second cell is connected to a second wordline adjacent to the target wordline. Illustratively, each cell of a pair of adjacent cells can be directly above or directly below its respective target cell.

The cell state information can include first cell state information obtained from the first cell and second cell state information obtained from the second cell. The first cell state information is indicative of a state of the first cell, and the second cell state information is indicative of a state of the second cell. For example, the first cell state information can include a V_T value indicative of the state of the first cell, and the second cell state information can include a V_T value indicative of the state of the second cell. Obtaining the cell state information can further include storing the cell state information for each pair of adjacent cells in a respective page buffer connected to the pair of adjacent cells via a bitline.

Any suitable number of reads can be performed on each cell of a pair of adjacent cells to obtain the cell state information. For example, during 4BCR, 3 reads can be performed on each adjacent cell of the pair. As yet another example, during 4BCR, 4 reads can be performed on each adjacent cell of the pair. As yet another example, during 4BCR, 5 reads can be performed on each adjacent cell of the pair. As yet another example, during 4BCR, 6 reads can be performed on each adjacent cell of the pair.

At operation **530**, a state information bin ("bin") assignment is performed. For example, the processing logic can, for each target cell of the set of target cells, determine a bin of a set of bins by applying a pre-defined operation to the first cell state information and the second cell state information of the respective pair of adjacent cells, and assign each target cell of the set of target cells to the respective bin. Each bin of the set of bins is assigned a respective read offset. Each target cell can be assigned to a respective bin by applying a pre-defined operation to the first cell state information and the second cell state information obtained from the respective pair of adjacent cells. The pre-defined operation is applied as part of a bin reduction scheme to reduce the number of bins, and thus the number of reads that will be performed with respect to the set of target cells.

In some embodiments, the pre-defined operation includes a linear combination of the first cell state information and the second cell state information. For example, the linear combination can be a sum of the first cell state information and the second cell state information obtained from the respective pair of adjacent cells. As another example, the linear combination can be a weighted sum of the first cell state information and the second cell state information. The bin reduction scheme described herein is not limited to linear combinations. For example, the bin reduction scheme can be extended to include other mathematical operations. Such mathematical operations can include the read level of the target cell, where different target cell read levels with corrective read can use a different mathematical operation with the state information.

The set of bins can include any suitable number of bins. For example, during 4BCR, if the number of reads per-

formed on each cell of the pair of adjacent cells is 3, then set of bins can include 7 bins. As another example, during 4BCR, if the number of reads performed on each cell of the pair of adjacent cells is 4, then the set of bins can include 9 bins. As yet another example, during 4BCR, if the number of reads performed on each cell of the pair of adjacent cells is 5, then the set of bins can include 11 bins. As yet another example, during 4BCR, if the number of reads performed on each cell of the pair of adjacent cells is 6, then the set of bins can include 13 bins.

At operation 540, the set of target cells is read. For example, the processing logic can cause each target cell of the set of target cells to be read using respective ones of the read level offsets. For example, a first read can be performed on the target cells of a first bin using the respective read level offset defined by the first bin. After the epilogue phase of the first read concludes, a second read can be performed on the target cells of a second bin, etc.

The number of reads performed on the set of target cells can be equal to the number of read level offsets, which is the same as the number of bins. For example, if the set of bins includes 7 bins, then the set of target cells can be read 7 times each with a different read level offset corresponding to a respective bin. As another example, if the set of bins includes 9 bins, then the set of target cells can be read 9 times each with a different read level offset corresponding to a respective bin. As yet another example, if the set of bins includes 11 bins, then the set of target cells can be read 11 times each with a different read level offset corresponding to a respective bin. As yet another example, if the set of bins includes 13 bins, then the set of target cells can be read 13 times each with a different read level offset corresponding to a respective bin. Further details regarding operations 510-540 are described above with reference to FIGS. 1A and 4 and will be described in further detail below with reference to FIGS. 6A-6D.

FIGS. 6A-6D illustrate example sets of bins that can be used to perform corrective reads with improved recovery from data retention loss, in accordance with some embodiments of the present disclosure. In each of the FIGS. 6A-6D, it is assumed that 4BCR is being implemented with respect to QLC cell implementation. It is further assumed in these examples that the bins are generated based on the linear combination $V_{T1} + V_{T2}$ with respect to the cell state information of each pair of adjacent cells.

FIG. 6A is a table 600A of an example set of bins generated using cell state information obtained by performing 3 reads with respect to each cell of each pair of adjacent cells (for a total of 6 adjacent cell read operations). As shown, states L0-L3 define a base bin 0 for each adjacent wordline, states L4-L7 define a base bin 1 for each adjacent wordline, states L8-L11 define a base bin 2 for each adjacent wordline, and states L12-L15 define a base bin 3 for each adjacent wordline. There are 16 total bins shown that are derived from the 4 base bins for each adjacent wordline. However, there are 7 effective "bin combinations" that each define a respective read level offset. Each effective bin combination represents a respective grouping of the bins. As shown, the 7 effective bin combinations include "bin combination 0" through "bin combination 6". The entry of each bin combination shown in the table 600A can be calculated as a sum of a respective pair of base bins (e.g., a sum of a base bin 3 entry for each adjacent cell results in an entry in bin combination 6).

Bin combination 0 includes combinations of L0-L3 for each of the adjacent wordlines. Bin combination 6 includes combinations of L12-L15 for each of the adjacent wordlines.

Bin combination 1 includes: (1) combinations of L4-L7 for adjacent wordline WL_{n-1} and L0-L4 of adjacent wordline WL_{n+1} and (2) combinations of L0-L4 for adjacent wordline WL_{n-1} and L4-L7 of adjacent wordline WL_{n+1} . Bin combination 5 includes: (1) combinations of L12-L15 for adjacent wordline WL_{n-1} and L8-L11 of adjacent wordline WL_{n+1} and (2) combinations of L8-L11 for adjacent wordline WL_{n-1} and L12-L15 of adjacent wordline WL_{n+1} .

Bin combination 2 includes: (1) combinations of L8-L11 for adjacent wordline WL_{n-1} and L0-L4 of adjacent wordline WL_{n+1} ; (2) combinations of L4-L7 for each of the adjacent wordlines; and (3) combinations of L0-L4 for adjacent wordline WL_{n-1} and L8-L11 of adjacent wordline WL_{n+1} . Bin combination 4 includes: (1) combinations of L12-L15 for adjacent wordline WL_{n-1} and L4-L7 of adjacent wordline WL_{n+1} ; (2) combinations of L8-L11 for each of the adjacent wordlines; and (3) combinations of L4-L7 for adjacent wordline WL_{n-1} and L12-L15 of adjacent wordline WL_{n+1} .

Bin combination 3 includes: (1) combinations of L12-L15 for adjacent wordline WL_{n-1} and L0-L4 of adjacent wordline WL_{n+1} ; (2) combinations of L8-L11 for adjacent wordline WL_{n-1} and L4-L7 of adjacent wordline WL_{n+1} ; (3) combinations of L4-L7 for adjacent wordline WL_{n-1} and L8-L11 of adjacent wordline WL_{n+1} ; and (4) combinations of L0-L4 for adjacent wordline WL_{n-1} and L12-L15 of adjacent wordline WL_{n+1} .

By using these 7 bin combinations, the number of bins can be reduced from 16 to 7 such that 7 target cell read operations can be performed (instead of 16). The total number of reads can then be about 32 reads (the 6 adjacent cell read operations, plus the 7 target cell read operations multiplied by 3.75 average reads per page on the target cell).

FIG. 6B is a table 600B of an example set of bins generated using cell state information obtained by performing 4 reads with respect to each cell of each pair of adjacent cells (for a total of 8 adjacent cell read operations). As shown, states L0-L2 define a base bin 0 for each adjacent wordline, states L3-L5 define a base bin 1 for each adjacent wordline, states L6-L8 define a base bin 2 for each adjacent wordline, states L9-L11 define a base bin 3 for each adjacent wordline, and states L12-L15 define a base bin 4 for each adjacent wordline. There are 25 total bins shown that are derived from the 5 base bins for each adjacent wordline. However, there are 9 effective "bin combinations" that each define a respective read level offset. Each effective bin combination represents a respective grouping of the bins. As shown, the 9 effective bin combinations include "bin combination 0" through "bin combination 8". The entry of each bin combination shown in the table 600B can be calculated as a sum of a respective pair of base bins (e.g., a sum of a base bin 3 entry for each adjacent cell results in an entry in bin combination 6).

Bin combination 0 includes combinations of L0-L2 for each of the adjacent wordlines. Bin combination 8 includes combinations of L12-L15 for each of the adjacent wordlines.

Bin combination 1 includes: (1) combinations of L3-L5 for adjacent wordline WL_{n-1} and L0-L2 of adjacent wordline WL_{n+1} and (2) combinations of L0-L2 for adjacent wordline WL_{n-1} and L3-L5 of adjacent wordline WL_{n+1} . Bin combination 7 includes: (1) combinations of L12-L15 for adjacent wordline WL_{n-1} and L9-L11 of adjacent wordline WL_{n+1} and (2) combinations of L9-L11 for adjacent wordline WL_{n-1} and L12-L15 of adjacent wordline WL_{n+1} .

Bin combination 2 includes: (1) combinations of L6-L8 for adjacent wordline WL_{n-1} and L0-L2 of adjacent wordline WL_{n+1} ; (2) combinations of L3-L5 for each of the

adjacent wordlines; and (3) combinations of L0-L2 for adjacent wordline WL_{n-1} and L6-L8 of adjacent wordline WL_{n+1} . Bin combination 6 includes: (1) combinations of L12-L15 for adjacent wordline WL_{n-1} and L6-L8 of adjacent wordline WL_{n+1} ; (2) combinations of L9-L11 for each of the adjacent wordlines; and (3) combinations of L6-L8 for adjacent wordline WL_{n-1} and L12-L15 of adjacent wordline WL_{n+1} .

Bin combination 3 includes: (1) combinations of L9-L11 for adjacent wordline WL_{n-1} and L0-L2 of adjacent wordline WL_{n+1} ; (2) combinations of L6-L8 for adjacent wordline WL_{n-1} and L3-L5 of adjacent wordline WL_{n+1} ; (3) combinations of L3-L5 for adjacent wordline WL_{n-1} and L6-L8 of adjacent wordline WL_{n+1} ; and (4) combinations of L0-L2 for adjacent wordline WL_{n-1} and L9-L11 of adjacent wordline WL_{n+1} . Bin combination 5 includes: (1) combinations of L12-L15 for adjacent wordline WL_{n-1} and L3-L5 of adjacent wordline WL_{n+1} ; (2) combinations of L9-L11 for adjacent wordline WL_{n-1} and L6-L8 of adjacent wordline WL_{n+1} ; (3) combinations of L6-L8 for adjacent wordline WL_{n-1} and L9-L11 of adjacent wordline WL_{n+1} ; and (4) combinations of L3-L5 for adjacent wordline WL_{n-1} and L12-L15 of adjacent wordline WL_{n+1} .

Bin combination 4 includes: (1) combinations of L12-L15 for adjacent wordline WL_{n-1} and L0-L2 of adjacent wordline WL_{n+1} ; (2) combinations of L9-L11 for adjacent wordline WL_{n-1} and L3-L5 of adjacent wordline WL_{n+1} ; (3) combinations of L6-L8 for each of the adjacent wordlines; (4) combinations of L3-L5 for adjacent wordline WL_{n-1} and L9-L11 of adjacent wordline WL_{n+1} ; and (5) combinations of L0-L2 for adjacent wordline WL_{n-1} and L12-L15 of adjacent wordline WL_{n+1} .

By using these 9 bin combinations, the number of bins can be reduced from 16 to 9 such that 9 target cell read operations can be performed (instead of 16). The total number of reads can then be about 42 reads (the 8 adjacent cell read operations, plus the 9 target cell read operations multiplied by 3.75 average reads per page on the target cell). This number is still less than the 66 reads that are performed during typical 4BCR.

FIG. 6C is a table 600C of an example set of bins generated using cell state information obtained by performing 5 reads with respect to each cell of each pair of adjacent cells (for a total of 10 adjacent cell read operations). As shown, states L0-L1 define a base bin 0 for each adjacent wordline, states L2-L4 define a base bin 1 for each adjacent wordline, states L5-L7 define a base bin 2 for each adjacent wordline, states L8-L10 define a base bin 3 for each adjacent wordline, states L11-L13 define a base bin 4 for each adjacent wordline, and states L14-L15 define a base bin 5 for each adjacent wordline. There are 36 total bins shown that are derived from the 6 base bins for each adjacent wordline. However, there are 11 effective "bin combinations" that each define a respective read level offset. Each effective bin combination represents a respective grouping of the bins. As shown, the 11 effective bin combinations include "bin combination 0" through "bin combination 11". The entry of each bin combination shown in the table 600C can be calculated as a sum of a respective pair of base bins (e.g., a sum of a base bin 3 entry for each adjacent cell results in an entry in bin combination 6). Each of the bin combinations can be determined in a similar manner as described above with reference to FIGS. 6A-6B.

By using these 11 bin combinations, the number of bins can be reduced from 16 to 11 such that 11 target cell read operations can be performed. The total number of reads in this case can be about 51 reads (the 10 adjacent cell read

operations, plus the 11 target cell read operations multiplied by 3.75 average reads per page on the target cell).

FIG. 6D is a table 600D of an example set of bins generated using cell state information obtained by performing 6 reads with respect to each cell of each pair of adjacent cells (for a total of 12 adjacent cell read operations). As shown, states L0-L1 define a base bin 0 for each adjacent wordline, states L2-L3 define a base bin 1 for each adjacent wordline, states L4-L6 define a base bin 2 for each adjacent wordline, states L7-L9 define a base bin 3 for each adjacent wordline, states L10-L11 define a base bin 4 for each adjacent wordline, states L12-L13 define a base bin 5 for each adjacent wordline, and states L14-L15 define a base bin 6 for each adjacent wordline. There are 49 total bins shown that are derived from the 7 base bins for each adjacent wordline. However, there are 13 effective "bin combinations" that each define a respective read level offset. Each effective bin combination represents a respective grouping of the bins. As shown, the 13 effective bin combinations include "bin combination 0" through "bin combination 11". The entry of each bin combination shown in the table 600D can be calculated as a sum of a respective pair of base bins (e.g., a sum of a base bin 3 entry for each adjacent cell results in an entry in bin combination 6). Each of the bin combinations can be determined in a similar manner as described above with reference to FIGS. 6A-6B.

By using these 13 bin combinations, the number of bins can be reduced from 16 to 13 such that 13 target cell read operations can be performed. The total number of reads in this case can be about 61 reads (the 12 adjacent cell read operations, plus the 13 target cell read operations multiplied by 3.75 average reads per page on the target cell).

The examples shown in FIGS. 6C-6D may require additional page buffers to implement, as compared to the examples shown in FIGS. 6A-6B. However, each of the examples shown in FIGS. 6A-6D can enable improved read performance with fewer reads per page.

FIG. 7 illustrates an example machine of a computer system 700 within which a set of instructions, for causing the machine to perform any one or more of the methodologies discussed herein, can be executed. In some embodiments, the computer system 700 can correspond to a host system (e.g., the host system 120 of FIG. 1) that includes, is coupled to, or utilizes a memory sub-system (e.g., the memory sub-system 110 of FIG. 1) or can be used to perform the operations of a controller (e.g., to execute an operating system to perform operations corresponding to the CR component 137 of FIG. 1). In alternative embodiments, the machine can be connected (e.g., networked) to other machines in a LAN, an intranet, an extranet, and/or the Internet. The machine can operate in the capacity of a server or a client machine in client-server network environment, as a peer machine in a peer-to-peer (or distributed) network environment, or as a server or a client machine in a cloud computing infrastructure or environment.

The machine can be a personal computer (PC), a tablet PC, a set-top box (STB), a Personal Digital Assistant (PDA), a memory cellular telephone, a web appliance, a server, a network router, a switch or bridge, or any machine capable of executing a set of instructions (sequential or otherwise) that specify actions to be taken by that machine. Further, while a single machine is illustrated, the term "machine" shall also be taken to include any collection of machines that individually or jointly execute a set (or multiple sets) of instructions to perform any one or more of the methodologies discussed herein.

The example computer system 700 includes a processing device 702, a main memory 704 (e.g., read-only memory (ROM), flash memory, dynamic random access memory (DRAM) such as synchronous DRAM (SDRAM) or RDRAM, etc.), a static memory 706 (e.g., flash memory, static random access memory (SRAM), etc.), and a data storage system 718, which communicate with each other via a bus 730.

Processing device 702 represents one or more general-purpose processing devices such as a microprocessor, a central processing unit, or the like. More particularly, the processing device can be a complex instruction set computing (CISC) microprocessor, reduced instruction set computing (RISC) microprocessor, very long instruction word (VLIW) microprocessor, or a processor implementing other instruction sets, or processors implementing a combination of instruction sets. Processing device 702 can also be one or more special-purpose processing devices such as an application specific integrated circuit (ASIC), a field programmable gate array (FPGA), a digital signal processor (DSP), network processor, or the like. The processing device 702 is configured to execute instructions 726 for performing the operations and steps discussed herein. The computer system 700 can further include a network interface device 708 to communicate over the network 720.

The data storage system 718 can include a machine-readable storage medium 724 (also known as a computer-readable medium) on which is stored one or more sets of instructions 726 or software embodying any one or more of the methodologies or functions described herein. The instructions 726 can also reside, completely or at least partially, within the main memory 704 and/or within the processing device 702 during execution thereof by the computer system 700, the main memory 704 and the processing device 702 also constituting machine-readable storage media. The machine-readable storage medium 724, data storage system 718, and/or main memory 704 can correspond to the memory sub-system 110 of FIG. 1.

In one embodiment, the instructions 726 include instructions to implement functionality corresponding to a CR component (e.g., the CR component 137 of FIG. 1). While the machine-readable storage medium 724 is shown in an example embodiment to be a single medium, the term "machine-readable storage medium" should be taken to include a single medium or multiple media that store the one or more sets of instructions. The term "machine-readable storage medium" shall also be taken to include any medium that is capable of storing or encoding a set of instructions for execution by the machine and that cause the machine to perform any one or more of the methodologies of the present disclosure. The term "machine-readable storage medium" shall accordingly be taken to include, but not be limited to, solid-state memories, optical media, and magnetic media.

Some portions of the preceding detailed descriptions have been presented in terms of algorithms and symbolic representations of operations on data bits within a computer memory. These algorithmic descriptions and representations are the ways used by those skilled in the data processing arts to most effectively convey the substance of their work to others skilled in the art. An algorithm is here, and generally, conceived to be a self-consistent sequence of operations leading to a desired result. The operations are those requiring physical manipulations of physical quantities. Usually, though not necessarily, these quantities take the form of electrical or magnetic signals capable of being stored, combined, compared, and otherwise manipulated. It has proven convenient at times, principally for reasons of common

usage, to refer to these signals as bits, values, elements, symbols, characters, terms, numbers, or the like.

It should be borne in mind, however, that all of these and similar terms are to be associated with the appropriate physical quantities and are merely convenient labels applied to these quantities. The present disclosure can refer to the action and processes of a computer system, or similar electronic computing device, that manipulates and transforms data represented as physical (electronic) quantities within the computer system's page buffers and memories into other data similarly represented as physical quantities within the computer system memories or page buffers or other such information storage systems.

The present disclosure also relates to an apparatus for performing the operations herein. This apparatus can be specially constructed for the intended purposes, or it can include a general purpose computer selectively activated or reconfigured by a computer program stored in the computer. Such a computer program can be stored in a computer readable storage medium, such as any type of disk including floppy disks, optical disks, CD-ROMs, and magnetic-optical disks, read-only memories (ROMs), random access memories (RAMs), EPROMs, EEPROMs, magnetic or optical cards, or any type of media suitable for storing electronic instructions, each coupled to a computer system bus.

The algorithms and displays presented herein are not inherently related to any particular computer or other apparatus. Various general purpose systems can be used with programs in accordance with the teachings herein, or it can prove convenient to construct a more specialized apparatus to perform the method. The structure for a variety of these systems will appear as set forth in the description below. In addition, the present disclosure is not described with reference to any particular programming language. It will be appreciated that a variety of programming languages can be used to implement the teachings of the disclosure as described herein.

The present disclosure can be provided as a computer program product, or software, that can include a machine-readable medium having stored thereon instructions, which can be used to program a computer system (or other electronic devices) to perform a process according to the present disclosure. A machine-readable medium includes any mechanism for storing information in a form readable by a machine (e.g., a computer). In some embodiments, a machine-readable (e.g., computer-readable) medium includes a machine (e.g., a computer) readable storage medium such as a read only memory ("ROM"), random access memory ("RAM"), magnetic disk storage media, optical storage media, flash memory components, etc.

In the foregoing specification, embodiments of the disclosure have been described with reference to specific example embodiments thereof. It will be evident that various modifications can be made thereto without departing from the broader spirit and scope of embodiments of the disclosure as set forth in the following claims. The specification and drawings are, accordingly, to be regarded in an illustrative sense rather than a restrictive sense.

What is claimed is:

1. A memory device comprising:

a memory array comprising:

- a target cell connected to a target wordline;
- a first wordline and a second wordline, each adjacent to the target wordline; and
- a pair of adjacent cells adjacent to the target cell, wherein the pair of adjacent cells comprises a first

27

cell connected to the first wordline and a second cell connected to the second wordline; and
 control logic, operatively coupled with the memory array, to perform operations comprising:
 causing a read operation to be initiated with respect to the set of target cell;
 obtaining first cell state information indicative of a first programming state for the first cell, wherein the first programming state corresponds to a first base state information bin associated with a first index value;
 obtaining second cell state information indicative of a second programming state for the second cell, wherein the second programming state corresponds to a second base state information bin associated with a second index value;
 identifying a target cell state information bin associated with a third index value determined based on the first index value and the second index value; and
 assigning the target cell to the target cell state information bin, wherein the target cell state information bin defines a read level offset for reading the target cell assigned to the target cell state information bin.

2. The memory device of claim 1, wherein the third index value is determined as a linear combination of the first index value and the second index value.

3. The memory device of claim 1, wherein causing obtaining the first cell state information comprises causing one or more read operations to be performed using one or more respective read voltages with respect to the first wordline.

4. The memory device of claim 1, wherein obtaining second cell state information comprises causing one or more read operations to be performed using one or more respective read voltages with respect to the second wordline.

5. The memory device of claim 1, wherein the operations further comprise causing the target cell to be read using the read level offset.

6. The memory device of claim 5, wherein:
 the target cell is comprised within a set of target cells that forms a pair of threshold voltage distributions of target cells associated with a center read level; and
 causing the target cell to be read using the read level offset comprises identifying a read level for reading the target cell by applying the read level offset to the center read level.

7. The memory device of claim 6, wherein the target cell state information bin corresponds to a threshold voltage sub-distribution of target cells, and wherein the read level has a magnitude below the threshold voltage sub-distribution.

8. A method comprising:
 causing, by a processing device, a read operation to be initiated with respect to a target cell connected to a target wordline of a memory device, wherein the target wordline is adjacent to a first wordline and a second wordline, wherein a pair of adjacent cells is adjacent to the target cell, and wherein the pair of adjacent cells comprises a first cell connected to the first wordline and a second cell connected to the second wordline;
 obtaining, by the processing device, first cell state information indicative of a first programming state for the first cell, wherein the first programming state corresponds to a first base state information bin associated with a first index value;
 obtaining, by the processing device, second cell state information indicative of a second programming state for the second cell, wherein the second programming

28

state corresponds to a second base state information bin associated with a second index value;
 identifying, by the processing device, a target cell state information bin associated with a third index value determined based on the first index value and the second index value; and
 assigning, by the processing device, the target cell to the target cell state information bin, wherein the target cell state information bin defines a read level offset for reading the target cell assigned to the target cell state information bin.

9. The method of claim 8, wherein the third index value is determined as a linear combination of the first index value and the second index value.

10. The method of claim 8, wherein obtaining the first cell state information comprises causing one or more read operations to be performed using one or more respective read voltages with respect to the first wordline.

11. The method of claim 8, wherein obtaining second cell state information comprises causing one or more read operations to be performed using one or more respective read voltages with respect to the second wordline.

12. The method of claim 8, further comprising causing, by the processing device, the target cell to be read using the read level offset.

13. The method of claim 12, wherein:
 the target cell is comprised within a set of target cells that forms a pair of threshold voltage distributions of target cells associated with a center read level; and
 causing the target cell to be read using the read level offset comprises identifying a read level for reading the target cell by applying the read level offset to the center read level.

14. The method of claim 13, wherein the target cell state information bin corresponds to a threshold voltage sub-distribution of target cells, and wherein the read level has a magnitude below the threshold voltage sub-distribution.

15. A non-transitory computer-readable storage medium comprising instructions that, when executed by a processing device, cause the processing device to perform operations comprising:
 obtaining first cell state information indicative of a first programming state for a first cell of a memory device, wherein the first cell is connected to a first wordline, and wherein the first programming state corresponds to a first base state information bin associated with a first index value;
 obtaining second cell state information indicative of a second programming state for a second cell of the memory device, wherein the second cell is connected to a second wordline, wherein the second programming state corresponds to a second base state information bin associated with a second index value, wherein the first cell and the second cell are each adjacent to a target cell connected to a target wordline, and wherein the first wordline and the second wordline are each adjacent to the target wordline;
 assigning the target cell to a target cell state information bin associated with a third index value determined based on the first index value and the second index value; and
 causing the target cell to be read using a read level offset defined by the target cell state information bin.

16. The non-transitory computer-readable storage medium of claim 15, wherein the third index value is determined as a linear combination of the first index value and the second index value.

17. The non-transitory computer-readable storage medium of claim 15, wherein obtaining the first cell state information comprises causing one or more read operations to be performed using one or more respective read voltages with respect to the first wordline.

5

18. The non-transitory computer-readable storage medium of claim 16, wherein obtaining second cell state information comprises causing one or more read operations to be performed using one or more respective read voltages with respect to the second wordline.

10

19. The non-transitory computer-readable storage medium of claim 15, wherein:

the target cell is comprised within a set of target cells forming a pair of threshold voltage distributions of target cells associated with a center read level; and
causing the target cell to be read using the read level offset comprises identifying a read level for reading the target cell by applying the read level offset to the center read level.

15

20. The non-transitory computer-readable storage medium of claim 19, wherein the target cell state information bin corresponds to a threshold voltage sub-distribution of target cells, and wherein the read level has a magnitude below the threshold voltage sub-distribution.

20

* * * * *

25